



A recent advances on autism spectrum disorders in diagnosing based on machine learning and deep learning

Hajir Ammar Hatim^{1,4} · Zaid Abdi Alkareem Alyasseri^{1,2,3} · Norziana Jamil⁵

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Abstract

Neurological disorders affect communication ability, social interaction, and a person's conduct. Early diagnosis and treatment of ASD during the early stages of a person's life may result in better outcomes and a higher quality of life for patients. Current methods of diagnosis are based on behavioral observations and interviews, which are subjective, time-consuming, and costly. EEG does not include invasive techniques, and it is a safe and painless way of measuring electrical activity in the brain. EEG signals may reflect neural differences and abnormalities related to ASD and serve as a potential biomarker for diagnosis. Due to the increase in prevalence, there has been an increased need to develop more sensitive and unbiased diagnostic methods for ASD. ML and DL are two sophisticated methods that researchers developed for detecting ASD by doing neural network analyses. The review paper incorporates the analysis of previous studies; more than 200 works have been analyzed from top publishers like Elsevier, IEEE, MDPI, and Springer, specifically related to EEG signal analysis and feature extraction techniques. It considers significant methods for ASD detection, including SVMs, CNN, and other models like KNN, ResNet50, and ANFIS. Other datasets central in these studies are KAU, BCIAUT-P300, and ADOS-2. The performance metrics adopted in this research include accuracy, sensitivity, and specificity. For example, the cubic SVM realized an accuracy of 95.8%, while the CNN models reached 95%. Other models, like ResNet50, achieved 99.39%, while ANFIS reached 98.9%. Sensitivity and specificity also showed varying scores across the methods, between 85 and 100%, indicating the high potential of these approaches in ASD diagnosis. Future studies could pay more attention to dataset representativeness improvements and do the clinical validation of these models for better generalization and relevance toward early diagnosis in ASD.

Keywords Autism · ASD · Machine learning · Deep learning · Diagnosis

1 Introduction

Autism spectrum disorder (ASD) encompasses a group of neurodevelopmental disorders that impact a person's interests, behaviors, communication abilities, social skills, and social interactions. Autistic symptoms and degrees of severity may range greatly; some people with the disorder have trouble communicating, interacting socially, and interpreting sensory information. On the other hand, others are very bright or averagely intelligent and shine in specific fields. One must keep in mind that autism is not a curable affliction. Early detection and diagnosis of ASD represent challenges globally. (Heunis et al. 2018; Aslam and Altaf 2019).

The CDC reports that the median age for diagnosing ASD may vary based on the type and severity of ASD, as well as the child's racial, ethnic, and socioeconomic background. Reports from 2021 indicated that 1 out of every 44 children were diagnosed with ASD. The prediction jumped to 1 in 36, only two years after the last study. The Autism and Developmental Disabilities Monitoring (ADDM) Network reports that the prevalence of ASD has increased by 317% since 2000. ASD affects roughly 4% of boys and 1% of girls. These figures are higher than previous estimates for the years 2000 to 2018. Research shows that ASD is more prevalent in males than females, with a rate of approximately 4 to 5 times higher. However, it is essential to note that these numbers may vary depending on the study and geographic location (Walensky et al. 2023; WHO 2023; McDougle et al. 2023) (Fig. 1).

Additional information regarding ASD includes that the disorder's prevalence exceeds that of AIDS, diabetes, and cancer put together; no known cause or cure exists for ASD. It is a lifelong developmental disability that usually appears before age three. No blood test or other medical diagnostic can identify ASD. However, early intervention is the most effective therapy (Reichow et al. 2018; McDougle et al. 2023; NIH 2023) (Fig. 2).

The diagnosis of ASD is based on the presence of persistent difficulties in two criteria: first, social for both communication and interaction and second, restricted repetitive and/or sensory behaviors or interests. There are three levels of severity for each criterion: the first, which calls for support; the second, which calls for more support; and the third, which calls for highly considerable help. The American Psychiatric Association's DSM-5 is the most generally used ASD diagnosis criterion. DSM-5 addresses ASD severity and accompanying problems such as intellectual disability, language difficulty, and seizures (Mayo Clinic 2018; CDC 2022). Before the DSM-5, ASD was divided into four subtypes (Rudy 2021):

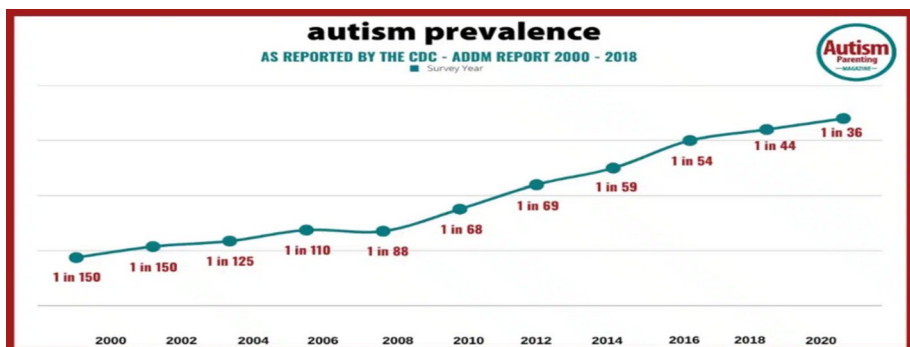


Fig. 1 Autism statistics since 2000 (Loftus 2023b, 2023b)

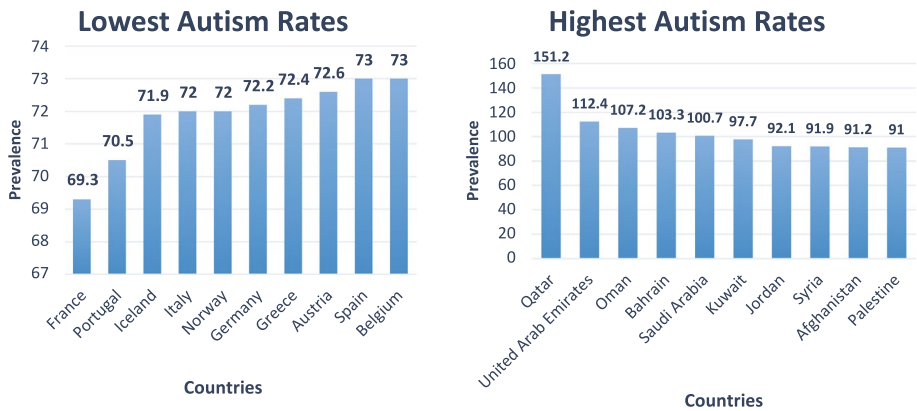


Fig. 2 The 10 Countries with the Autism Rates (WorldPopulationReview 2023)

1. Pervasive Developmental Disorder—Not Otherwise Specified (PDD-NOS).
2. Asperger's Syndrome (AS).
3. Autistic Disorder.
4. Childhood Disintegrative Disorder (CDD).

These subtypes differed in the onset, severity, and pattern of symptoms, but they are now considered to be part of the same spectrum of ASD ((CDC) 2022).

The methods used in hospitals to diagnose autism are psychological and behavioral assessments (CDC 2023b). New techniques and technologies are developed and implemented to improve the early identification and intervention of ASD, and the essential methods used to diagnose and treat autism include facial features, expression analysis, facial landmark detection, robot-accessed interaction, eye tracking, eye contact training, brain MRI and EEG signals measurement, and genetic testing. These methods may help to detect ASD in younger children, especially those who do not show obvious signs or symptoms, and provide them with appropriate and timely support and treatment (Loftus 2023b, 2023b, page number).

Traditional methods of ASD diagnosis often rely on behavioral observation and clinical interviews, which, although effective, may be quite cumbersome and costly, let alone subjective. These methods are also challenging for early diagnosis, especially in young children who may not manifest clear-cut behavioral signs. The challenges of analyzing such complex data signals for finding patterns and making predictions with a high degree of accuracy can be overcome by using the techniques of Machine Learning and Deep Learning. An automated approach would reduce subjectivity and hasten diagnosis with enhanced precision in ASD detection. It reviews current research on the applications of ML and DL in ASD diagnosis, highlighting main methods and datasets to guide future research toward improving early and accurate identification of ASD (Peya et al. 2020; Sharabash and Elghaish 2024).

Although the current review focuses on machine and deep learning methods based on EEG, other modalities have also demonstrated promising results. For example, (Mujeeb Rahman and Monica Subashini 2022a, 2022b) employed deep neural networks using the QCHAT questionnaire and achieved an accuracy of 98%. Eye-tracking data has likewise been explored as a potential biomarker using CNN and boosting algorithms (Kanhirakada-

vath and Chandran 2022). Furthermore, facial-image-based DNN models have been proposed for ASD screening (Mujeeb Rahman and Subashini 2022). These studies illustrate the potential of integrating behavioral, visual, and physiological signals into AI-based ASD diagnostic frameworks.

1.1 Fundamentals of EEG and data acquisition

Electroencephalography (EEG) is a non-invasive method investigating the brain's electrical activity. The EEG records electrical signals originating from brain activity through electrodes placed on the scalp, further detecting very small electrical impulses produced by neuronal activity. This technique offers valuable information about the functional state of the brain since it records brainwave patterns associated with various cognitive and neurological conditions, such as autism spectrum disorder (ASD). Electrode placement is normally done by the international 10–20 electrode placement system so that recordings are consistent between different studies. The 10–20 system is shown in Fig. 3. Normally, data acquisition is performed in an externally noiseless environment; signals are filtered to remove interference and artifacts. These are then digitized for further processing. EEG provides high temporal resolution, is valuable for ASD research, and is relatively low-cost and non-invasive. Therefore, it forms an important tool for studying brain function in research and the clinic (Hu and Zhang 2019).

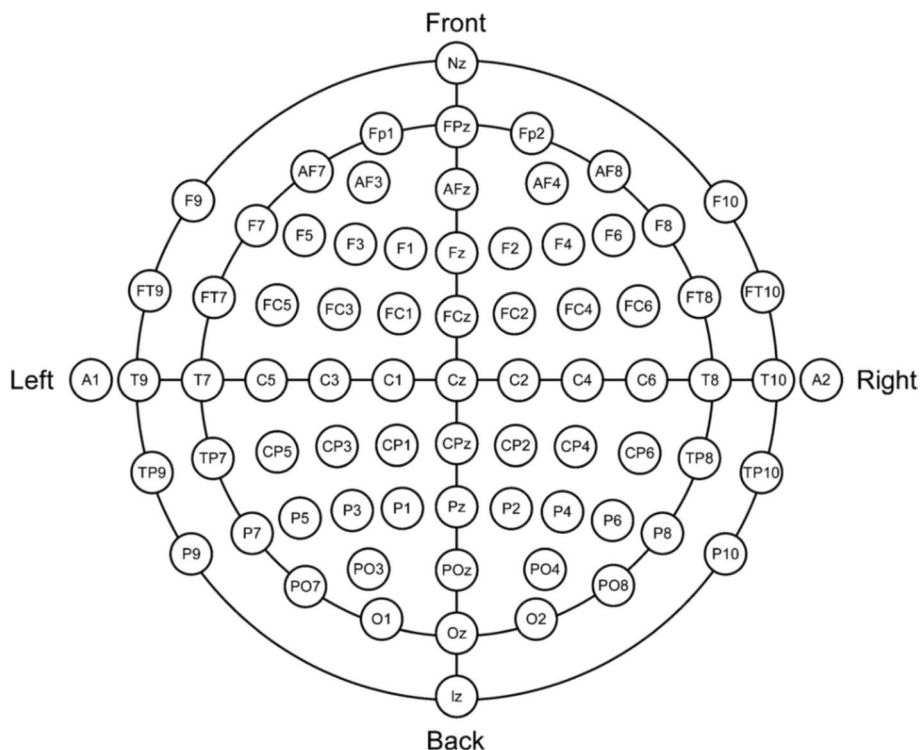


Fig. 3 International 10–20 System for electrode placement (Hu & Zhiguo, 2019)

Although EEG holds great promise as a non-invasive diagnostic tool for ASD, several challenges are ahead. Biological signals like EEG are easily susceptible to noise and interference, so the results cannot be all that reliable. Besides, this measures a relatively lower temporal resolution than other imaging techniques. Limitations include susceptibility to biological noise and signal interference, which may make results unreliable. Besides, there is a relatively low temporal resolution compared to other imaging techniques. To address such limitations, several researchers suggested integrating EEG data with other biomarkers, such as neuroimaging results, genetic information, or behavioral assessments, to improve the validity of the diagnosis of ASD in general (Yun [2024](#); Sun et al. [2023](#); Dickinson et al. [2024](#)).

1.2 Diagnose and treat autism using EEG signals

Identifying ASD at an early stage can lead to better developmental outcomes through early intervention. Early electroencephalography (EEG) shows great promise in infants for the prediction of ASD diagnosis in the future and identification of the mechanisms for neural brain processes that are associated with the disorder (Peck et al. [2021](#)). New technologies and methods are being developed and implemented to improve the early identification and intervention of ASD using EEG signal measurements in Artificial Intelligence (AI). The brain produces waves of electrical activity known as EEG signals reflected from the cerebral cortex, which can be used to evaluate its functional health for the brain. These signals can differ between individuals with and without ASD in brain activity and connection. Analyzing, classifying, and predicting the risk and severity of ASD based on EEG data is made possible by machine learning and deep learning methods in artificial intelligence with up to 95% accuracy as early as three months of age (W.J. Bosl, Tager-Flusberg, and Nelson [2018a, 2018b](#)).

Numerous studies have proposed novel algorithms and models to enhance the quality and variability of EEG signals. These models are employed to recognize the subtypes of ASD by analyzing EEG patterns. By leveraging these methods and technologies, it is possible to create cost-effective, non-intrusive, and accurate tools to detect and address ASD in its early stages (Tawhid et al. [2021a](#); Esqueda-Elizondo et al. [2022](#)).

EEG signals help detect differences in brain activity between individuals with and without ASD. They can also predict the risk and severity of ASD based on EEG features. EEG signals have several advantages over other brain imaging methods, such as their low cost, non-invasive nature, and higher temporal resolution. Furthermore, the effects of ASD treatments and interventions may be monitored using EEG data. Thus, EEG data may serve as a valuable instrument for the early detection and treatment of ASD (Peng et al. [2021](#); Gurau, Bosl, and Newton [2017](#); Li et al. [2023](#)).

1.3 Diagnose and treat autism using brain-computer interface

The new technology promising to diagnose and treat ASD is the brain-computer interface (BCI), which affects how people communicate, interact, and behave. BCI technology uses electroencephalography (EEG) signals to measure and analyze the brain activity of individuals with ASD. It can give feedback or stimulation to modulate brain function and behavior. BCI technology can help detect the emotional state, attention level, and cognitive abilities



Fig. 4 The System components of BCI

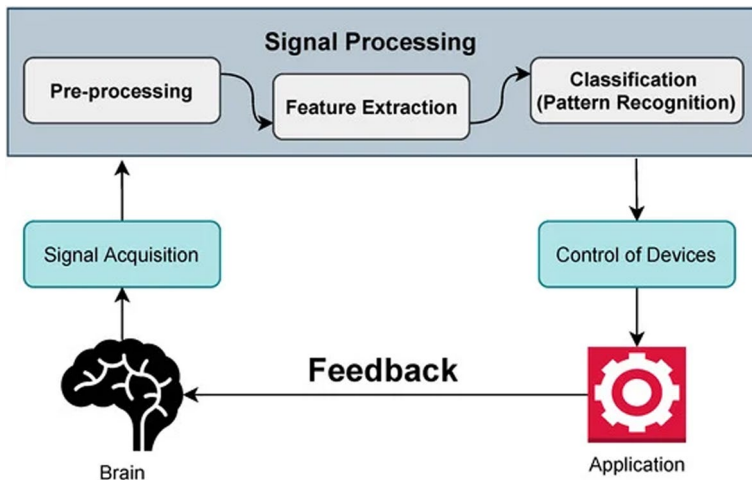


Fig. 5 The fundamental structure of the system BCI (Mridha, et al. 2021b, 2021b)

of individuals with ASD. By analyzing EEG features, it can predict the risk and severity of ASD (Vilchis et al. 2023; Zaman et al. 2022). BCI technology can also improve the social and communication skills, executive functions, and adaptive behaviors of individuals with ASD. It provides personalized and engaging interventions like games, music, or neurofeedback. BCI technology has the potential to provide non-invasive and objective instruments for the detection and intervention of ASD, which can improve the quality of life for individuals with ASD and their families while keeping the cost low (Han 2021; Ravindranathan, Tommy, and R 2020; Mridha et al. 2021b, 2021a) (Figs. 4, 5).

This article thoroughly overviews recent studies that might help medical professionals and scientists diagnose autism spectrum disorder. Springer, Elsevier, IEEE, and Wiley were among the esteemed journals that contributed over 250 articles that were evaluated. For the benefit of future researchers, this review study may provide helpful details on popular and publicly accessible ASD datasets. Also, it will encourage more studies into utilizing EEG for early identification of autism, which will help researchers in the field. There are some of the questions this article will try to address:

Q1: What ML and DL processes are used to diagnose ASD?

Q2: Which countries publish the most autism research?

Q3: Can a publicly available dataset be applied for ASD detection?

Q4: What are the most commonly used filters in pre-processing signal-processing techniques applied to raw EEG data?

Q5: What role might EEG Feature Extraction play in diagnosing ASD?

Q6: What are the most relevant metrics when evaluating ML and DL methods?

Q7: Is it available any application end-to-end for ASD diagnosis?

The following sections comprise the rest of this review: Sect. 2 outlines the methodology used to select studies related to autism spectrum disorder. Section 3 presents results and discussion, which provides a comprehensive analysis of techniques used for ASD evaluation. Section 3.1.1 focuses on works related to machine learning (ML) approaches, and Sect. 3.1.2 presents the research on deep learning (DL) approaches. Section 3.1.3 presents a hybrid approach that combines both ML and DL. Available ASD datasets are provided in Sect. 3.2. Section 3.3 presents EEG pre-processing, followed by EEG Features Extraction in Sect. 3.4. The review concludes with Sect. 4, which includes information on the most commonly used ML and DL techniques for ASD evaluation. Finally, Sect. 5 recommends future work.

2 Methodology for selecting studies related to autism spectrum disorder

This section provides a methodology for selecting related studies on autism spectrum disorder. The most relevant keywords were used to pick the studies: ‘autism,’ ‘feature,’ and ‘EEG.’ Digital databases were utilized, and papers published in English, such as Elsevier, IEEE, Springer, MDPI, Wiley, and others, were retrieved. Our methodology consisted of five phases. The first step was to look for research related to autism spectrum disorder (ASD). Two hundred twenty-eight scientific papers were recovered in September 2023, including research on feature analysis and EEG for autism. In the first stage, we combined the three terms—‘autism,’ ‘feature,’ and ‘EEG’—using the ‘AND’ operator. We removed duplicate papers in the second phase, which disqualified 62 papers. In the third phase, we screened the selected scientific papers based on their titles and abstracts and discarded publications with themes beyond the scope of our research. After rejecting 76 articles, we were left with only 90 papers. In the fourth phase, we skimmed each document to ensure it was within the designated scope and eliminated 17 articles. Finally, we included 73 publications on ASD characteristics and EEG. Figure 6 shows the methodology of the paper selection process. All three phases employed the same qualifying criterion (Fig. 7).

Thus, the reviewed studies have primarily adopted machine learning models like Support Vector Machines and deep learning models like Convolutional Neural Networks. It also included the BCIAUT_P300 dataset, which adopted different ML techniques and other datasets such as KAU and ADOS-2. These datasets consisted of EEG recordings and behavioral assessments. Their sample sizes were very different, with some studies using larger samples, such as BCIAUT_P300, while others focused on smaller cohorts. Second, preprocessing, including noise filtering, normalization, and artifact removal, has been an important step in improving the quality of the EEG signal. In this respect, the steps discussed here ensure the data becomes clean and adequate for analysis, enhancing the models’ accuracy. The methods used for feature extraction included wavelet transform, Fast Fourier Transform, and other techniques that emphasized relevant EEG patterns for its diagnosis.

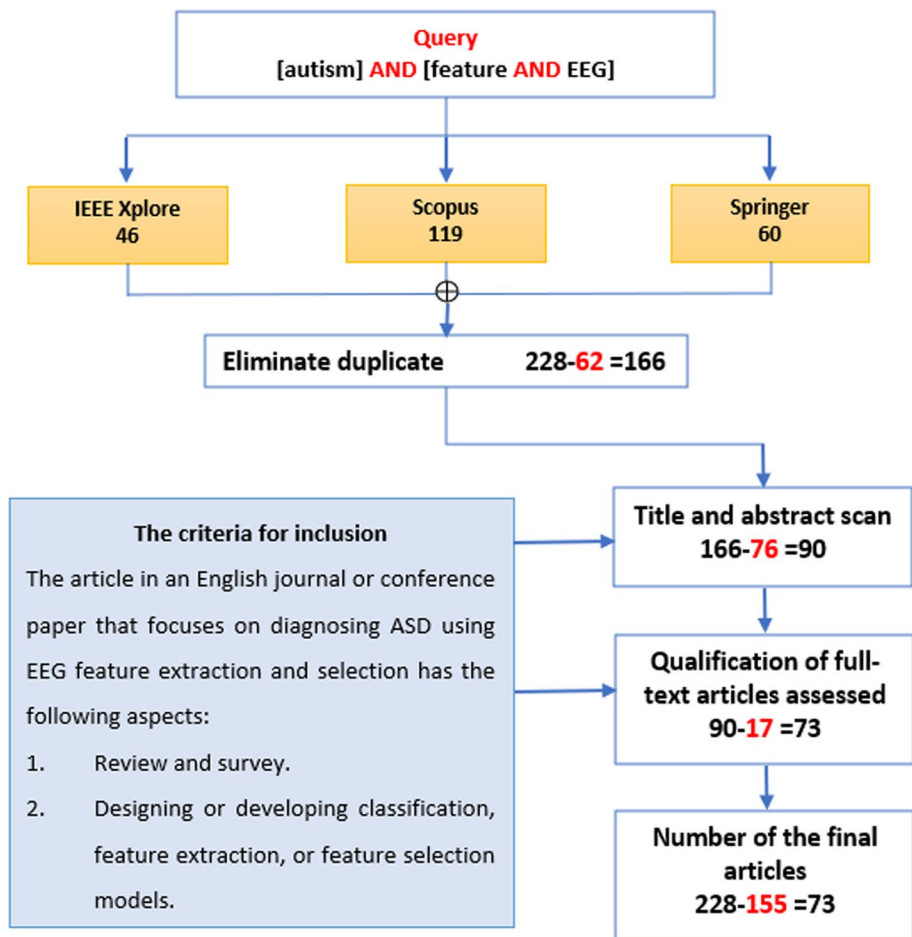


Fig. 6 Methodology of selected studies involving the query and inclusion criteria

3 Result and discussion

3.1 Comprehensive finding and analysis of autism spectrum disorder techniques

Autism spectrum disorder (ASD) prevalence has increased in the past decade. Currently, the only method used to diagnose ASD is a qualitative behavioral assessment by professionals using internationally recognized descriptive scale standards, and it is still the gold standard for diagnosing ASD (CDC 2023a; Abdulhay et al. 2018; Mayo Clinic 2018). In recent years, there has been a rise in the application of machine-learning techniques for detecting ASD. The increase in ASD has resulted in the need for more accurate and objective detection methods. Machine learning (ML) and deep learning (DL) techniques have been employed to enhance the performance of traditional ASD detection and prediction methods. Although these methods have yet to be widely adopted or validated, they have shown promising outcomes (Radhakrishnan et al. 2021; Meridian et al. 2022; Bahathiq et al. 2022).

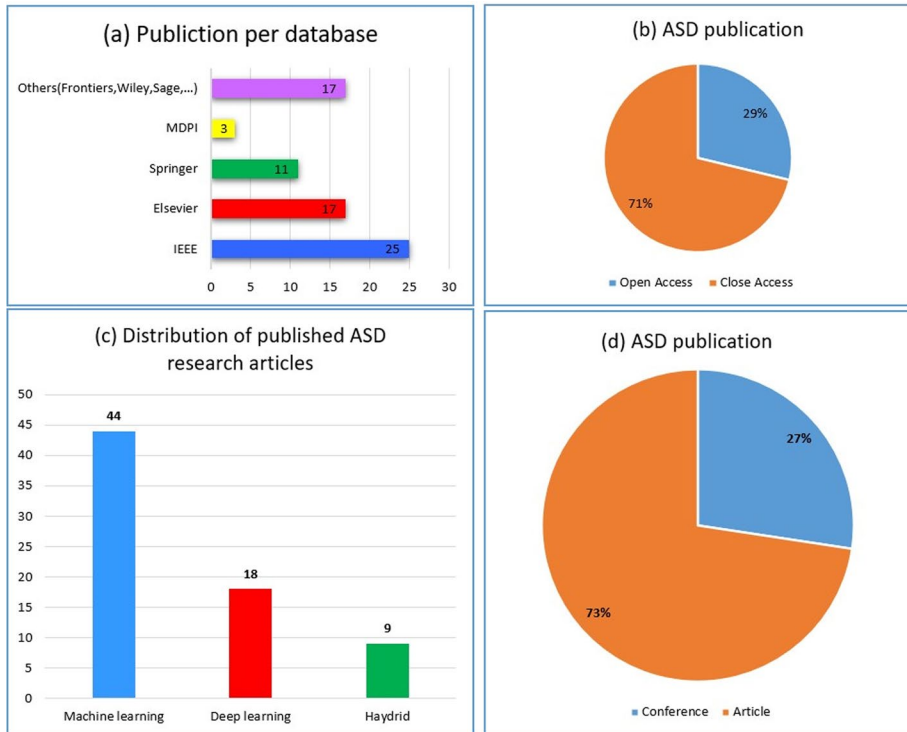


Fig. 7 Statistics publications on ASD

This section analyzes 73 previous studies on the diagnosis or detection of ASD using electroencephalography (EEG) signals. The analysis revealed that ML, DL, or hybrid approaches that combine ML and DL with feature extraction techniques effectively diagnose or detect ASD. The assessment measures showed varying effectiveness of these methods, ranging from 63 to 100%.

Different methods are used to detect autism spectrum disorder (ASD). In Sect. 3.1.1, we covered machine learning, and in Sect. 3.1.2, we discussed the use of various deep-learning methods for ASD detection. Additionally, Sect. 3.1.3 describes hybrid approaches that combine ML and DL for ASD detection.

3.1.1 Machine learning approaches

EEG signals have been analyzed and classified using Artificial Intelligence (AI) to significant effect. Machine Learning (ML) has played a substantial role in enhancing these results. ML is a branch of artificial intelligence that develops algorithms and statistical models. These models allow computers to learn and make decisions independently without explicit programming. Several ML categories include supervised, unsupervised, semi-supervised, and reinforcement learning (Patel et al. 2023, page number).

Medical signal and image analysis have been significantly improved by machine learning systems that automatically find and extract relevant features for the diagnosis of ASD,

which is much better than traditional diagnostic systems (Muñoz-Saavedra et al. 2020, page number).

According to previous studies that used machine learning approaches shown in Figs. 8, 9 to diagnose ASD by using electroencephalography (EEG) signals, the best results from these studies include:

3.1.1.1 Neural networks, artificial neural networks, and probabilistic neural networks (Abdulhay et al. 2018) The results from a primary dataset showed that using a second-order difference plot (SODP) feature for neural network (NN) classification into ASD and neurotypical groups leads to 94.4% accuracy and 100% sensitivity. There is a significant difference in the area values of SODP among ASD children and those with typical development.

(Ibrahim et al. 2018) Used a fusion of Discrete wavelet transform (DWT), Shannon entropy, and artificial neural networks (ANNs) techniques to detect autism, resulting in an accuracy of 99.71 while an overall accuracy of classification problems for (normal vs. epilepsy vs. autism) multi-channel up to 94.6%. EEG applications utilize an emerging non-linear data analysis technique called Recurrence quantification analysis (RQA). Another study (Grossi et al. 2020, page number) used quantitative time series features of two studies conducted previously; the first study has twelve features extracted from the C3–C4 time series (Grossi et al. 2017, page number), and the second study extracted 36 features from the C3–C4 time (Grossi et al. 2019) representing EEG signals. Based on these features, the best ANN, acting as a classifier, could accurately predict whether a case was autistic or typical in Study 1 with 100% accuracy. Study 2 was able to differentiate between other neurological and psychiatric disorders with an accuracy of 94.95%.

(Pham et al. 2020) proposed a method for diagnosing autism spectrum disorder (ASD) using EEG signals. They used the Locality Sensitivity Discriminant Analysis technique (LSDA) to simplify the EEG features. These features were based on the signal's power spectrum and entropy. They then selected the five most significant features using the sta-

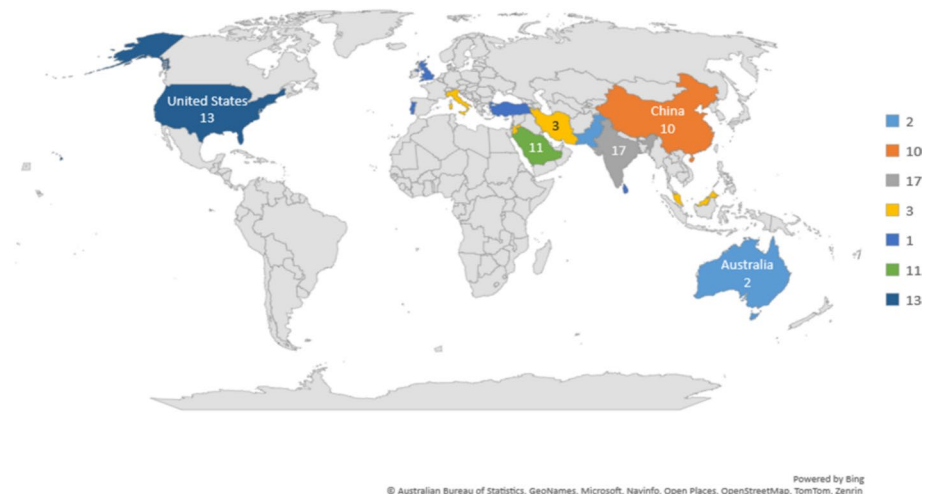


Fig. 8 Recent studies of countries targeting autism 2018–2023

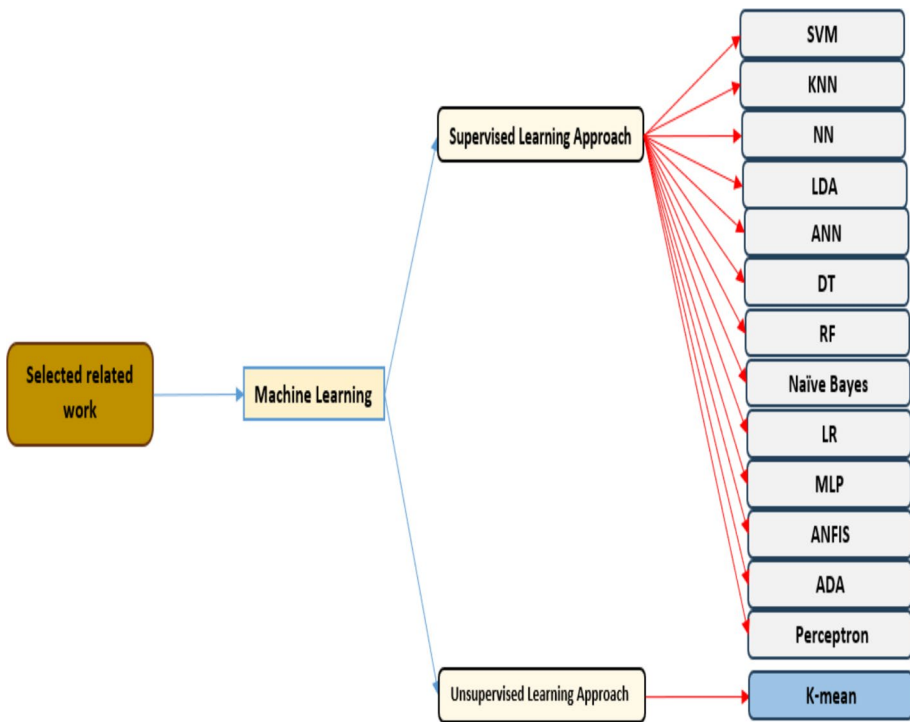


Fig. 9 Categorizing of related works using machine learning approaches for ASD

tistical method known as Student's t-test and fed them into different classifiers. Among the classifiers, the probabilistic neural network (PNN) achieved the highest accuracy of 98.70% in distinguishing ASD from typical cases.

3.1.1.2 Support vector machine The study (Heunis et al. 2018) applied the support vector machine (SVM) classifier with Recurrence Quantification Analysis (RQA) features, and they presented results indicating an accuracy of 92.9% and 100% sensitivity to demographic features (RQA, age, and gender).

In a study conducted by (Abdolzadegan et al. 2020), they explored the use of SVM in classifying data. They explored the use of SVM in categorizing data. The study found that features such as entropy, discriminant function analysis (DFA), Lyapunov exponent (LE), and synchronization likelihood (SL) significantly improved classification accuracy when used with Power Spectrum (PS), Wavelet Transform (WT), and Fast Fourier Transform (FFT). The study achieved a classification accuracy of 90.57% and a sensitivity of 99.91% for SVM. Non-linear features were extracted from run-length matrix-based features.

(Alturki et al. 2020) the best results were achieved when DWT was combined with entropy or local binary pattern (LBP), and either SVM or artificial neural network (ANN) was used for classification. They worked for both single-channel and multi-channel data. In the three-class classification (neurotypical vs. epilepsy vs. autism) for single and multi-

channel data, SVM achieved an overall accuracy of almost 99.9%, while ANN achieved 97% accuracy.

Significant results of the proposed ASD detection using BCI were introduced by (Rechal, Kumar, and Khaleelulla 2021), utilizing Variational Mode Decomposition (VMD) and Predictor Importance Estimates (PIE) techniques to extract features resulted in a high accuracy rate of 98.03%. VMD is a signal fragmentation method that adapts to optimize real-time signals through frequency-shifting properties and Hilbert transforms. On the other hand, PIE was used to select the most suitable features from the extracted ones.

The brain connectivity network can be characterized quantitatively by graph-theoretic parameters using three different phase-locking value (PLV) approaches: trial-averaged PLV, averaged trial-averaged PLV, and time-points-averaged PLV. These features were used to train machine learning classifiers to classify ASD. The cubic support vector machine classifier achieved excellent results with 95.8% accuracy, 100% sensitivity, and 92% specificity (Alotaibi and Maharatna 2021).

An automatic technique based on waves (75–250 Hz) entropy was used to classify the EEG of individuals with ASD and typically developed individuals. The method employed a scaled conjugate gradient (SCG) based neural network to classify entropy values. The overall and best testing accuracies achieved were 98.9% and 95.0%, respectively (Abdulhay et al. 2023).

(Divya et al. 2023), Used to classify high- and low-functioning autism in a kernel-based fine Gaussian SVM. The DWT technique extracted frequency band-specific absolute power, relative power, and derived parameters such as theta-to-beta and ratio theta-to-alpha ratio, and then took relevant features were statistically analyzed. The classifier achieved an average 99.9% accuracy, outperforming other kernel methods.

3.1.1.3 K-nearest neighbor Said (Grossi et al. 2019), the EEG processing approach has shown that it can differentiate between the computed EEG of Italian with typically developing and ASD children with almost perfect accuracy by using the Multi-Scale Ranked Organizing Map/Implicit Function As Squashing Time (MS-ROM/IFAST). He aimed his study to test the system's ability to identify ASD from other neuropsychiatric diseases (NPD). The results indicated a predictive capability ranging from 93 to 97.5%, with the top machine learning system achieving 95.0% accuracy worldwide for the K-Nearest Neighbor (KNN) (classifier in 1 out of 3 class distinctions).

In a study by (Alturki et al. 2021), they used Common Spatial Pattern (CSP) and LBP to extract features based on band power, energy, and entropy. They found that the CSP+LBP+KNN combination had the highest accuracy in diagnosing ASD, with an average classification accuracy of approximately 98.46%.

3.1.1.4 Adaptive neuro-fuzzy inference system (Kumar et al. 2019) extracted 11 features: wavelet entropy (WE), approximate entropy (ApEn), relative wavelet energy (RWE), root mean square (RMS), Hurst exponent (H), correlation dimension (CD), principal component analysis (PCA), higher order spectra (HOS), detrended fluctuation analysis (DFA), skewness (SK), and kurtosis (KU) together, used as input to the adaptive neuro-fuzzy inference

system (ANFIS) classifier enhanced by a hybrid bat (HBA) algorithm to detect autism in children and achieved accuracy of 98.9%, 97% to sensitivity and specificity.

The study (Bakheet and Maharatna 2021) utilized various techniques to extract signal features. These techniques included multivariate empirical mode decomposition (MEMD), sample entropy (SampEn), and standard linear measurements such as maximum, minimum, and standard deviation. Statistical tests were performed on these features to create input vectors for three classifiers—SVM, KNN, and Discriminant Analysis (DA). The experiment recorded signals responding to three facial expressions—happy, fearful, and neutral. The results showed that the system achieved a prediction accuracy of 100% for the happy stimulus dataset, indicating highly reliable results.

(Alhassan and Soudani 2022) He presented a study that uses sensors based on Haar wavelet transform and dynamics features to detect autism in children at an early stage. The experimental results showed that the proposed approach has an accuracy of 93%, a sensitivity of 86%, and a specificity of 100%.

Based on these works, we found that EEG can detect abnormalities in the neural system associated with ASD. It can also be used with various signal processing and machine learning techniques to identify features differentiating ASD from other neurological and psychiatric conditions, such as SODP, DWT, Shannon entropy, RQA, LSDA, etc. With the help of different classifiers such as NN, ANN, PNN, SVM, KNN, and others, SVM was used more and achieved high results, as shown in Fig. 10, with a percentage of machine learning approaches with ASD.

EEG can achieve high accuracy and sensitivity in classifying ASD. The prefrontal and temporal cortices are the areas where EEG can detect and measure dysfunctions in neuronal regulation in ASD. Due to its noninvasive, easy-to-use, and cost-effective nature, EEG can be a valuable and affordable test for diagnosing ASD (Alhassan, Soudani, and Almusallam 2023).

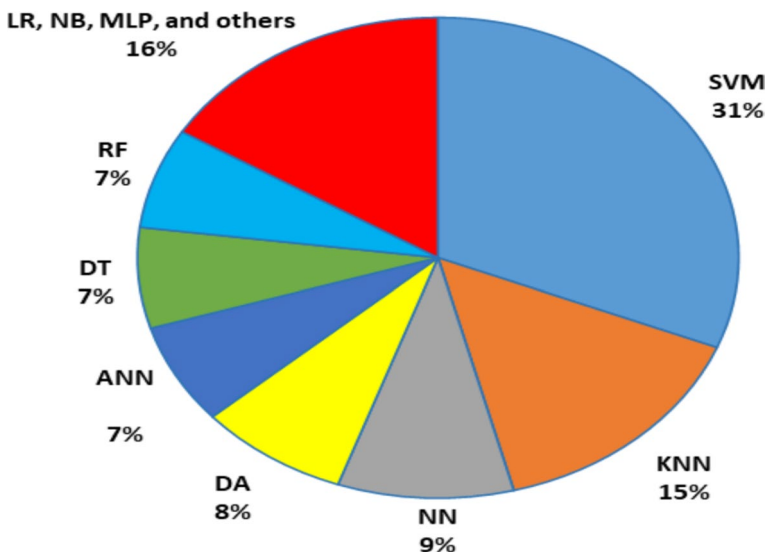


Fig. 10 Percentage of machine learning approaches with ASD

3.1.2 Deep-learning approaches

Deep learning (DL) is a form of machine learning that utilizes several layers of non-linear information processing to extract features, transform data, analyze patterns, and classify information. The feature extraction process allows deep learning to handle different data types, including unstructured data. As a result, deep learning models tend to have better performance metrics when dealing with such datasets (Radhakrishnan et al. 2021). The Convolutional Neural Network (CNN) primary algorithms in DL detect ASD in Sect. 3.1.2.1, which is covered, RNN covered in Sect. 3.1.2.2, and SNN and GNN in Sect. 3.1.2.3.

According to previous studies that used deep learning algorithms shown in Fig. 11 to diagnose ASD, the best results from these studies include:

3.1.2.1 Convolutional neural network According to (Alyasseri et al. 2022), Deep learning is a term used to describe multi-layered artificial intelligence neural networks (ANN). Pattern recognition has significantly improved in recent years due to deep neural networks outperforming classical methods. The most commonly used type of deep neural network is the Convolutional Neural Network (CNN). This type of network involves a linear mathematical operation between matrices, which is convolution. CNNs consist of multiple layers, such as convolutional, non-linear, pooling, and fully connected, each with its own set of parameters. These networks have achieved exceptional results in machine learning problems, particularly in applications that deal with image data, such as computer vision, natural language processing, and image recognition. This article provides how the CNNs are used to detect Autism Spectrum Disorder (ASD) and how they collaborate.

Assuming a solid understanding of computers and Artificial Neural Networks (ANNs), this paper defines the parameters that impact CNN efficiency. Figure 12 provides an example of a deep-learning approach for CNN.

(Mohi-Ud-Din and Jayanthi 2023) CNN is suggested for identifying ASD using scalogram images derived from EEG data. The EEG data obtained from people with Autism and typical control participants was divided into segments, and the Continuous Wavelet Trans-

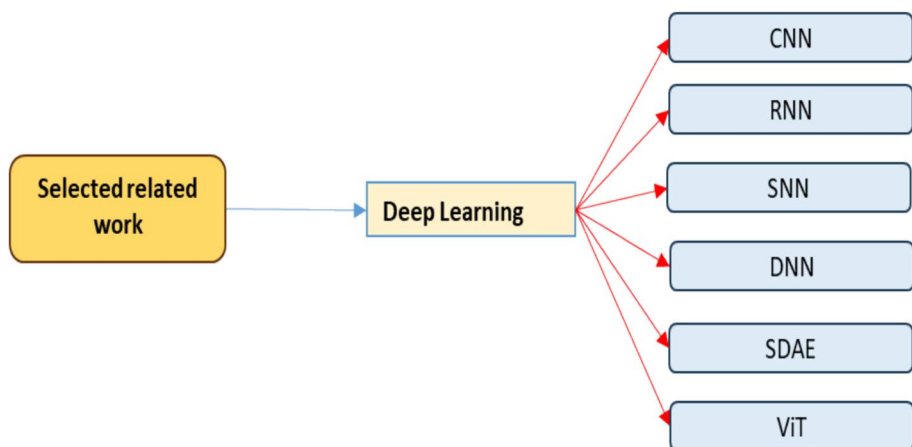


Fig. 11 Categorizes the related works that utilize DL approaches for ASD

form (CWT) technique was used to generate scalograms from these segments. The model underwent training using scalogram pictures and attained a precision rate of 77%.

Another study (Mohi-Ud-Din and Jayanthi 2021b) was conducted to identify individuals with ASD using scalograms. This technique provides a visual display of the wavelet transform of a signal, indicating both the time and frequency components. The researchers used this method to create channel-wise EEG data and then trained pre-existing neural networks such as Google Net and Squeeze Net. Both neural networks accurately distinguished ASD subjects from typical controls with a 75% and 82% success rate, respectively.

(Radhakrishnan et al. 2021) was conducted to compare the effectiveness of various deep convolutional architectures in detecting ASD. These architectures can automatically extract features and classify them with high accuracy. The study assessed the efficiency of various advanced deep learning models, including AlexNet, DenseNet, SqueezeNet, ShuffleNet, VGGNet, InceptionNet, and ResNet. The models were evaluated based on different performance metrics. The study revealed that ResNet50 achieved the highest accuracy, with an average accuracy rate of 81%. Furthermore, ResNet50 demonstrated excellent precision of 92%, sensitivity of 91%, F1-score of 91%, and specificity of 100%.

(Mohi-Ud-Din and Jayanthi 2021a) A CNN is used in conjunction with EEG signals to detect ASD. The CNN comprises convolution layers, which extract features, and max pooling layers, which perform feature reduction. This CNN was trained using EEG signals from both individuals with ASD and neurotypical subjects. The model achieved an accuracy of 92%, and its classification accuracy increased with more subjects in the training data.

(Peya et al. 2020) used a CNN to identify ASD using an EEG-based investigation. The researchers employed Pearson's correlation coefficient to transform single EEG channel

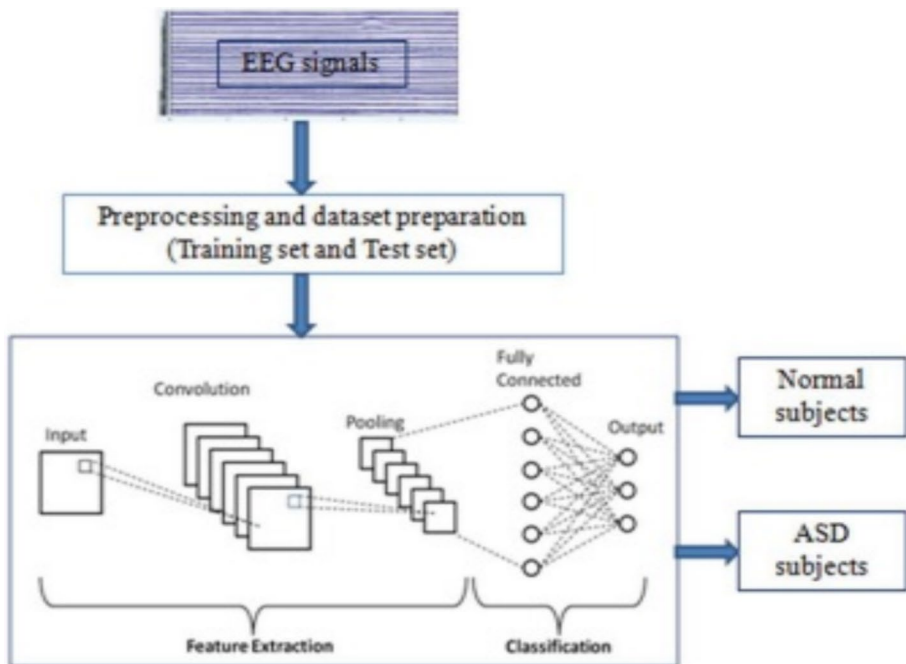


Fig. 12 Workflow of CNN overall (Mohi-Ud-Din and Jayanthi 2021a)

data into two-dimensional (2D) images, then classified using the CNN model residual neural network (ResNet). The experimental results demonstrated a remarkable accuracy of 100%.

To distinguish between ASD and typical development (TD) using EEG signals, (Wadhera and Mahmud 2022) conducted a study that developed a two-layer visual graph convolutional network (VGCN). The VGCN achieved a classification accuracy of 93.78%, which was higher than other techniques such as support vector machines (89.52%), deep neural networks (78.21%), convolutional networks (83.88%), and graph networks (86.45%). Furthermore, other evaluation metrics, such as the Mathews correlation coefficient, recall, and F1 score, also showed comparable results.

The study of (Mayor Torres et al. 2023) was the first to use the feature-relevance computing approach to discern emotions from EEG data in persons with and without ASD. The researchers developed the RemOve-And-Retrain (ROAR) approach to facilitate the retrieval of exceptionally relevant characteristics from pre-trained deep neural networks. The researchers assessed the ROAR algorithm's effectiveness in Face Emotion Recognition (FER), which has practical applications in the study of ASD. A CNN was used to train EEG data and evaluate the significance of EEG characteristics evoked by Facial Emotion Recognition (FER) in persons with and without ASD. The study used prior neuroscience and ASD research results and demonstrated a classification accuracy of 93.4%.

(Ari et al. 2022) has developed a new automated method that uses EEG recordings to detect ASD. The technique combines the Douglas-Peucker (DP) algorithm, a sparse coding-based feature mapping approach, and CNNs. The study used wavelet transform to extract EEG rhythms and Extreme Learning Machines (ELM) to increase the dataset. The study results showed that the method had an F1-score of 99.19% and an accuracy, sensitivity, and specificity of 98.88%, 100%, and 96.4%, respectively.

(Tawhid et al. 2023) introduced the automatic framework by using EEG to detect four neurological disorders, common among them is autism; the system diagnosis was developed by computer-aided diagnosis (CAD) depending on CNN and transformed spectrogram images into 2D based on time–frequency employed short-time Fourier transform (FFT). This method achieved an accuracy of 99.39% compared to AlexNet and ResNet50, standard CNN models.

According to (Bouallegue et al. 2020), the authors highlight the significance of the pre-processing step in identifying and processing the most informative frequency bands for a specific neurological disorder. The study employed a combination of gated-recurrent unit (RNN-GRU) and finite and infinite impulse response (FIR and IIR) filters, which helped extract pertinent features from complex EEG data, thereby improving the diagnosis process. The authors applied their system to autism and achieved an impressive accuracy rate of 99.5% with the King Abdulaziz University (KAU) dataset, using Independent Component Analysis (ICA) for feature extraction and CNN for classification.

3.1.2.2 Recurrent neural network A Recurrent Neural Network (RNN) is an artificial neural network capable of processing sequential data, such as electroencephalography (EEG). Researchers have utilized RNNs along with EEG data to detect Autism Spectrum Disorder (ASD) and other mental states and disorders (Pan et al. 2022; Zhang et al. 2022).

The use of RRN is essential in the development of six custom filters for the preprocessing step. These filters, which include Finite Impulse Response (FIR), Infinite Impulse Response

(IIR), notch, band-pass, high-pass, and low-pass filters, operate on each channel's epoch level of the EEG data. Once the signal is filtered, it trains an RNN that employs Gated Recurrent Units (GRUs). The trained network selects and applies the appropriate filter in a specific order, which helps to decrease the Mean Squared Error (MSE) at the epoch level and improve classification accuracy. Therefore, the proposed intelligent network RNN, which uses FIR and IIR filter design, is a dynamic filtering approach that optimizes classification accuracy (Bouallegue et al. 2020).

Various well-known techniques have been used to improve current algorithms, including SVM, LDA, CNN, and Long-Short Term Memory (LSTM). However, the study by (Zhao et al. 2020a, 2020b, 2020c) introduced a revolutionary RNN approach that utilized various pre-processing methods. This unique RNN solution achieved an accuracy score of 92.59%, 0.61 percentage points higher than the previously best-reported method during the IFMBE SC. Additionally, the solution produced a compatible accuracy of 89.92% using a conventional 80%-20% initial separation method.

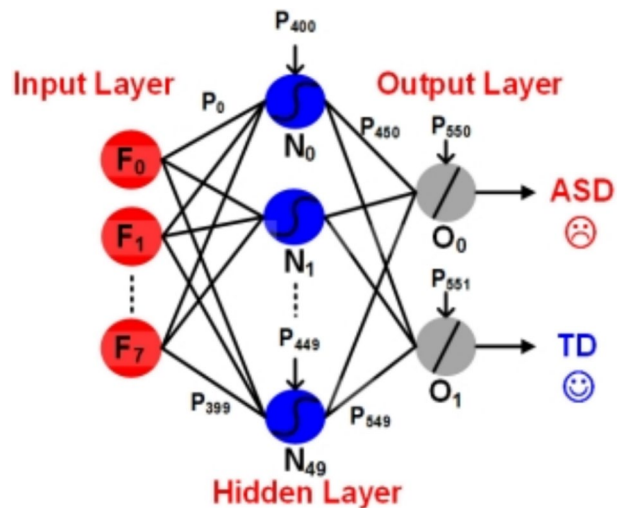
(Manoharan and Radhakrishnan 2023) Aimed to examine the neural dynamics of 20 children with ASD and 20 typical development (TD) using non-linear EEG signal analysis. Recurrence Quantification Analysis (RQA) was utilized to uncover the underlying non-linear EEG dynamics. The cosine distance metric was selected for its ability to retrieve information. Each computational combination of RQA measures and the responding channel was used to identify the best biomarker. The resulting features were then input into the RNN called the Bidirectional Long Short-Term Memory (BiLSTM) network, and channel-wise classification accuracy was tested. The T3 and T5 channels with FAN neighborhood selection and the cosine distance metric were the most suitable combinations, achieving a classification accuracy of 91.86%.

3.1.2.3 Other deep-learning approaches Neural networks can be categorized into different types based on their architecture. One such type is a Shallow Neural Network (SNN), a fully connected neural network without multiple hidden layers. The SNN uses backpropagation to optimize the weights and biases during the learning process by comparing the desired output with the actual output. The output is classified as ASD and TD. Neural networks are becoming increasingly popular in various fields due to their ability to learn quickly from small datasets (Aslam et al. 2021). The architecture of the SNN used to classify ASD is depicted in Fig. 13.

(Aslam et al. 2021) presented an ASD classification processor based on an EEG for wearable contexts and long-term monitoring. The algorithm uses the Artix-7 FPGA, four EEG electrodes with just 26 K lookup tables and 15 K flip-flops. The system employs Katz fractal dimension features with an efficiency of 38% and a kurtosis value of 54%. Using the SNN architecture, classification is accomplished with a high accuracy of 85.5% at 2.25 ms and 8.62 μ W of power and latency.

Han et al. (2022) It developed a framework that combines EEG and eye-tracking data to diagnose ASD in children. The framework uses a two-step process that involves learning and combining features from both modalities. A stacked denoising autoencoder (SDAE) was used for this purpose. The MMSDAE model, created using this framework, achieved the best performance in terms of evaluation metrics. It showed an accuracy of 95.56%, a sensitivity of 92.5%, and a specificity of 98%.

Fig. 13 The architecture of the SNN (Aslam et al. 2021)



The issue of restricted data size can be overcome, and model performance can be improved by using self-supervised learning. To fully utilize the data structure's temporal, spectral, and spatial characteristics, it is essential to consider its long-range correlations. A method for self-supervised EEG based on the vision transformer (ViT) has been proposed. The ViT can now learn internal EEG characteristics from a large dataset. Over a billion EEG data points from more than 4500 subjects were used to pre-train the model, which was later used to identify children with ASD. The experiment demonstrated that the pre-trained model outperformed the supervised learning model with an accuracy of 91.3%. This study has shown how universal temporal-spectral-spatial data structures can effectively train large deep-learning models and how the representation can be improved. (Chen, Gaoxiang, and Li 2023).

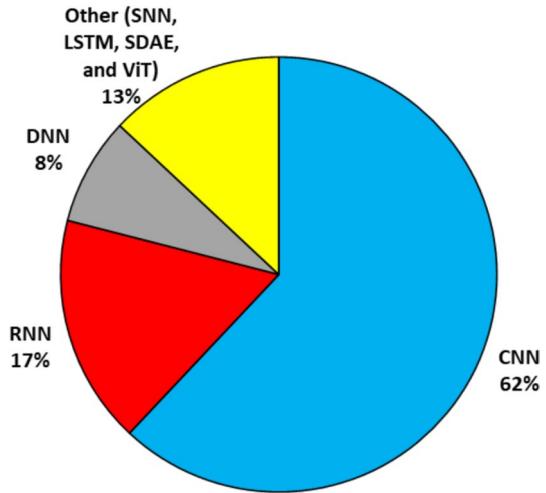
Based on these works, we found that EEG can detect abnormalities in the neural system associated with ASD. It can also be used with various signal processing and deep learning techniques to identify features differentiating ASD from other neurological and psychiatric conditions, using classifiers such as CNN, RNN, DNN, SNN, and others; CNN was used more and achieved high results as shown in Fig. 14, with a percentage of deep learning approaches with ASD.

3.1.3 Combining machine and deep learning in a hybrid approach

The most prominent results that both techniques used machine learning and deep learning were:

(Bouallegue et al. 2020), It was found that using complex deep neural network classifiers is crucial compared to simpler supervised and unsupervised classification algorithms. These simpler algorithms included support vector machines (SVM), k-nearest neighbor algorithm (KNN), k-means clustering, and convolutional neural network (CNN). The study also tested various feature extraction techniques, such as principal component analysis (PCA), Common Spatial Pattern (CSP), and independent component analysis (ICA). Results showed that KNN can produce comparable results (87.44%) when using CSP features. However,

Fig. 14 Percentage of deep learning methods with ASD



CNN with ICA provides the best classification performance with an impressive result of 99.50%. On the other hand, CSP and PCA produced results of 92.40% and 96.12%, respectively. Consequently, the study concludes that CNN with ICA is the most appropriate classifier design tool.

Tawhid et al. (2021b) An efficient diagnostic framework was developed to identify ASD automatically using time–frequency spectrogram images of EEG signals. The pre-processed signals were transformed into two-dimensional spectrogram images using the Short-Time Fourier Transform. ML approaches were then used to extract estimated images and select textural and essential features using PCA. These features were used as inputs to 6 different ML classifiers for classification. In contrast, DL techniques were tested using three different CNN models. The DL models showed the best accuracy of 99.15%, while the ML methods showed an accuracy of 95.25%.

(Sivasaravana Babu et al. 2023) A machine learning and deep learning algorithm, heuristic-concatenated feature classification algorithm (H-CFCA), was used to detect autism and epileptic seizures. EEG signals were pre-processed by breaking the waves down into smaller waves (5 frequency bands) using a Daubechies wavelet filter (DB4) and reducing dimensions using Singular Value Decomposition (SVD). The features were then extracted using Heuristic Independent Component Analysis (H-ICA) and fed into the H-CFCA system. The system consisted of a Support Vector Machine (SVM) and Bi-Stack Long Short-Term Memory (BiS-LSTM). The algorithm achieved impressive results with an accuracy of 99.51%, a sensitivity of 98.86%, and an F1 score of 99.32%.

Visu et al. (2023) used adaptive thresholding, discrete wavelet transforms (DWT), machine learning methods to support SVM, and ICA to pre-process, decompose, and separate signals of EEG. The features extracted from the heuristic-ICA technique fed the deep learning algorithm Memory-based CNN (M-CNN) to train and classify the EEG signals. The proposed method outperforms existing techniques with 97.6% accuracy and 94.2% sensitivity while addressing the gradient vanishing problem (Fig. 15).

3.2 Available autism spectrum disorder datasets

Neurological disease diagnosis, particularly autism spectrum disorder, faces challenges due to size, diverse data, and techniques (Jui et al. 2023).

Obtaining appropriate datasets for training, testing, and evaluating techniques for diagnosing autism spectrum disorder (ASD) can be a challenging task. To assist with this process, we have compiled a list of commonly used ASD datasets, each with unique features. Table 1 will find a comprehensive overview and instructions on accessing the datasets. We gathered these datasets from reputable digital publisher databases such as Springer, Elsevier, IEEE Xplore, MDPI, Frontiers, and Wiley using relevant keywords such as autism, EEG, and feature diagnosis. Still, almost all of the datasets were unavailable online, and some links to the dataset or the code were provided to request it because of secret information. Based on Table 1, the authors have provided and contributed datasets with varying sizes and types to support ASD using EEG signals.

Gathering a large amount of reliable data at a low cost is crucial yet difficult in various fields. Measuring and recording the brain's electrical activity through EEG on people diagnosed with ASD by attaching electrodes to their scalp or brain surface can provide valuable insights into their brain activity and connectivity. It can be a cost-effective method for diagnosis, prognosis, intervention, and research purposes (Wadhera, Bedi, and Sharma 2023a).

Several steps must be followed to gather a dataset of EEG signals for ASD research. The suitable EEG device and electrodes should be selected based on the research question, experimental design, budget, and device availability. Various types are available, such as wet, dry, semi-dry, wireless, wired, active, and passive, including devices like g.tec, Emotiv, Muse, NeuroSky, and OpenBCI. Next, a sample of individuals with ASD should be recruited, typically with TD individuals as a control group, ensuring they meet inclusion and exclusion criteria, obtain informed consent, and get ethical approval. The sample size and participant characteristics should be determined to prepare them for EEG recording. The process involves cleaning the scalp, applying conductive gel or water, placing electrodes, checking impedance, calibrating the EEG device, and following manufacturer instructions. EEG signals should be presented and recorded from stimuli to elicit emotional, cognitive, or behavioral responses from participants with ASD. Data should be stored securely in formats like (.edf, .set, .mat, or .csv), labeled with participant ID, session number, condition, and date, and backed up with encryption for data integrity and confidentiality.

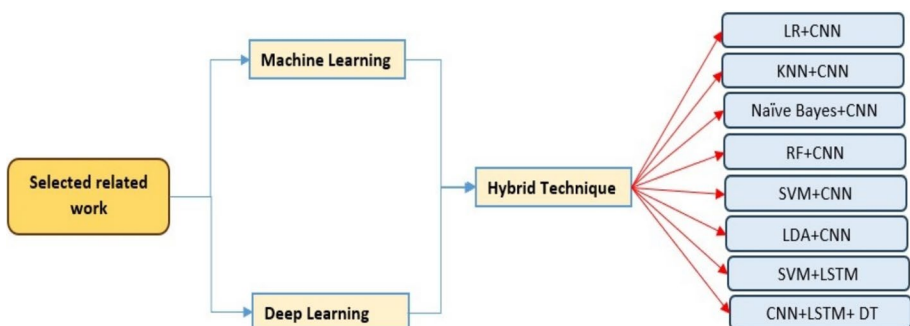


Fig. 15 Categorizing of related works using Hybrid approaches for ASD

Table 1 Available autism spectrum disorder datasets

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
KAU (Alhaddad et al., 2012)	EEG recordings	The dataset has 18 files: 8 for disorders and 10 for standard data. Eight boys (aged 10–16) are in the disorder group, while ten boys (aged 9–16) are in the standard group without any neurological disorder. The EEG recordings were taken at rest using active electrodes, an active digital EEG amplifier, and a recording system from BCI2000 at a sampling rate of 256 Hz. The recording system included g.tec EEG-cap, 16 Ag/AgCl electrodes, g.tec GAMMAbox, g.tec USBamp, and BCI2000. A bandpass filter (0.1–60 Hz) and a notch filter at 60 Hz were used to filter and digitize the data	✓	(Kumar et al. 2019; Bouallegue et al. 2020b; Alturki et al. 2020; Tawhid et al. 2021; Muñoz-Saavedra et al. 2020; Wadhera and Mahmud 2022; Aslam et al. 2022; Thirumal and Thangakumar 2022; Tawhid et al. 2023; Wadhera et al. 2023; Torres et al. 2023)	Accuracy: 99.9%, sensitivity: 100%, specificity: 97%, AUC: 99.8	https://malhaddad.kau.edu.sa/Pages-BCI-Datasets-En.aspx	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
ADOS-2-ASD-EEG (Jayarathna et al. 2019)	EEG recordings spectrogram image	The EEG dataset was obtained from 6 male and two female individuals affected by 8 ASD, aged between 5 and 15 years, and four male and five female typically developing (TD) individuals aged between 6 and 16 years. The recordings were made during the ADOS-2 administration and resting state using a 32-electrode wireless Live-Amp EEG system with FCz as the reference channel. The raw EEG signals were processed using various tools such as Independent Component Analysis, channel re-referencing, noise removal, ocular and artifact removal, and interpolation. The data was then divided into baseline with 15,000 samples and start, middle, and end epochs, each with 45,000 samples	✓	(Jayawardana, Jaime, and Jayarathna 2019; Aslam et al. 2021; Wadhera, Bedi, and Sharma 2023a; Qaysar Mohi Ud Din and Jayanthi 2023)	Accuracy: 90%, sensitivity: 93%, specificity: 92.4%	Not available online	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
BCIAUT-P300 (Simões et al. 2020)		The BCIAUT-P300 was a dataset from the 15th Mediterranean Conference on Medical, Biological Engineering, and Computing from a clinical trial that trained individuals diagnosed with an autism spectrum disorder to recognize and respond to social cues. An electroencephalography (EEG) system called G.Nautilus was used to record the data. Eight active electrodes were placed on the scalp in specific places (C3, Cz, C4, CPz, P3, Pz, P4, and Poz). The data, sampled at 250 Hz, was divided into seven sessions. These sessions ensured a comprehensive analysis of P300 variations over time, with training and test subsets. Fifteen individuals participated in a task that simulated real-world social interactions within a virtual environment. They were asked to identify and highlight one of eight objects	✓	(Adama, Schindler, and Schmid 2020; Peketi and Dhok 2023; de Arancibia et al. 2020; Li et al. 2020; Miladinović et al. 2020; Patel et al. 2023; Zhao, Yu, et al. 2020; Krzemi et al. 2020; Santamaria-Vázquez et al. 2020; Borra et al. 2020)	Accuracy: 93.16%, sensitivity: 94.65%, specificity: 91.40%, F1 score: 92.27%, precision: 94%,	BCIAUT_P300 (kaggle.com)	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Baseline, with-voice, and without-voice datasets (Abdolzadegan et al. 2020)	EEG recordings (numerical)	The EEG signal recording stage involves selecting channels and protocols, recording data from two channels at 256 Hz, and recording under three conditions: baseline, with and without voice. Data is recorded twice from each child, with a two-month interval between recordings, using a ten-channel FlexComp Infniti Encoder and BioGraph Infniti 6.0 software	✓	(Abdolzadegan et al. 2020)	Accuracy: 90.57%, sensitivity: 99.91%	Not available online	11/29/2023
VSM-EEG (Grossi et al. 2017)	EEG recordings	The dataset consists of 35 Italian subjects, including 15 with ASD and 10 typically developing, aged between 7 and 14. A psychiatrist diagnosed ASD using the ADOS scale. The EEG signals were acquired in Villa Santa Maria on a Micromed device, using 19 electrodes according to the 10–20 system. The recording rate was 256 Hz; recordings lasted for 3 min in a resting state	✓	(Grossi et al. 2020, Peya et al. 2020)	Accuracy: 100%, sensitivity: 94.95%, specificity: 100%	Not available online	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary	Techniques	Implementation	Available online	Accessed
ASD-EEG (Dong et al. 2021)	EEG recordings	The dataset contains EEG data of 175 Chinese children aged 3 to 6. It was used to evaluate the stability and accuracy of a reconstruction CNN model, a new approach to diagnosing ASD by analyzing EEG features. The study was conducted in four regions: Beijing, Ningxia, Guangdong, and Guangxi	✓	(Dong et al. 2021)	Accuracy: 100%, sensitivity: 94.95%, specificity: 100%, 0.82, RMSE: 2.9	Not available online	11/29/2023
Eye State Detection via EEG (Rajesh et al. 2015)		The dataset is a collection of EEG data from a single subject, recorded for 117 s with eyes open and closed. It includes 14,980 time points, is divided into 140 batches based on eye state changes, and is labeled binary (0 for open and 1 for closed). The data follows the International 10–20 System standard for EEG electrode placement	✓	(Mengucci et al. 2021)	Accuracy: 73.5%, ROC AUC: 79%	Home—UCI Machine Learning Repository	11/29/2023
Kaggle (Amaral et al. 2017)	EEG recordings	This dataset includes EEG signals from 8 channels (C3, Cz, C4, CPz, P3, Pz, P4, POz) collected from 23 autistic and 23 healthy children	✓	(Roopa Rechal, Rajesh Kumar, and Ebraheem Khaleelulla 2021)	Accuracy: 98.03%, sensitivity: 100%, specificity: 99.6%, precision: 99.2% F1-score: 99.6% G_Mean: 99.5%	Complete EEG dataset (kaggle.com)	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
SAFARI	EEG recordings	The Simons Foundation for Autism Research Initiative collected EEG signals from 57 people, including 16 control subjects aged one year to six months to 14 years and 41 ASD subjects aged ten months to 21.3 years. They filtered the signals using high-pass, notch, and band-pass filters and transformed them into 40,000 scalograms using the continuous wavelet transform in MATLAB	✓	(Mohi-Ud-Din and Jayanthi 2021a, 2023)	Accuracy: 92.2%, sensitivity: 94.5%, specificity: 90.5%, precision: 68%, Prevalence 0.34	Not available online	11/29/2023
EFE-ERP (Jamal et al. 2014; Apicella et al. 2013)	EEG recordings	The dataset comprises EEG data from 24 participants displaying neutral, happy, and fearful facial expressions. A 128-channel device was used to record the data, which was then filtered utilizing a high-pass filter of 0.5 Hz and a low-pass filter of 50 Hz. The filtered data was segmented into 1000 ms epochs, and any signals exceeding 200 μ V were rejected as artifacts. Data was collected from eight channels located in the fusiform gyrus cortex area	✓	(Bakheet and Maharatna 2021)	Accuracy: 94.4%, TRN: 100%, TRP: 89%	Not available online	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
ASD-EEG (Bajestani et al. 2019)	EEG recordings	The dataset comprises EEG data from 60 children between 4 and 8 years old. Thirty of these children have ASD, while the remaining 30 have typical development. The data was recorded using a 10-channel device and divided into 20-s segments. Moreover, this data was collected under an ethics license from the Medical Sciences University of Mashhad	✓	(Bajestani et al. 2019; Ghoreishi et al. 2021)	Accuracy: 81.96%, sensitivity: 66.66%, specificity: 94.11%	Not available online	11/29/2023
VEP Signals Empirical SSVEP Dry-EEG Datasets (Aznan et al. 2019)	VEP recordings	The dataset contains both actual and simulated VEP signals obtained from 30 individuals with ASD presented with varying contrasts. The data was captured via a 128-channel device at a sampling rate of 1000 Hz. Six electrodes from the occipital region were selected from these signals and converted to a standard 10–10, 81-channel system using the EEGLAB toolbox	✓	(Sheela and Puthankattil 2022)		Not available online	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
CHB-MIT (Ary L. Goldberger. et al., 2000)	EEG recordings	The data set includes EEG recordings from 22 pediatric subjects with intractable seizures at Children's Hospital Boston. The subjects were monitored after discontinuing anti-seizure medication to evaluate their suitability for surgical intervention. The data set is grouped into 23 cases, with most files containing 23 EEG signals. The International 10–20 system was used for recording, and the data set includes 182 seizures of various types and durations	✓	(Sivasarana Babu et al. 2023)	Accuracy: 99.51%, Sensitivity: 98.86%, F1 score: 99.32%	CHB-MIT Scalp EEG Database v1.0.0 (physionet.org)	11/29/2023
DEAP (Koelstra et al. 2012)	EEG recordings	A total of 120 music video clips were rated by 14–16 volunteers based on their level of arousal, valence, and dominance. Additionally, 32 volunteers were selected to watch 40 videos while their EEG and physiological signals were recorded. Furthermore, 22 of these participants also recorded a frontal face video to understand the effects of music on brain function	✓	(Aslam and Altaf 2019, Habeeb et al. 2019, Aslam et al. 2022)	Accuracy: 63%	https://paperswithcode.com/dataset/deap	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
ShapHT ASD-EEG (Yin et al. 2022)	EEG recordings	Beijing Normal University used eight channels to collect the dataset, which contained EEG signals from 113 kids, including 54 with autism spectrum disorder and 59 who were still developing. The dataset is a three-order tensor pre-processed using a bandpass filter, adaptive artifact detection, downsampling, DFT, and tensor construction	✓	(Yin et al. 2022)	Accuracy: 82.44%	GitHub—findings/ShapHT-Feature-Selection: Code: Adaptive feature selection with shapely and hypothetical testing: A case study of EEG feature engineering	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Post visual stimuli (face pictures) data (Mayo Clinic College of (Medicine and Rochester 2016; Catarino et al. 2013)	EEG recordings	The dataset consists of EEG data collected from 32 scalp electrodes that capture frequencies ranging from delta to gamma bands. The data has a high sampling rate of 1000 Hz and a lowpass filter of 50 Hz, which enables recording a broad range of EEG frequencies. The data is segmented into epochs that are time-locked to the onset of face stimuli and have a duration of 400 ms. The dataset includes ten face processing channels: T7, F7, F8, TP8, T8, P7, F8, TP7, and F8. The dataset is divided into two groups, ASD and TD, with varying mean and standard deviations	✓	(Alhassan and Soudani 2022)	Accuracy: 93%, sensitivity: 86%, specificity: 100%, Positive PPV: 100%, Negative Predictive Value (NPV): 100%	Not available online	11/29/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Han EEG (Han et al. 2017)	EEG recordings	The dataset was obtained from 88 ASD and 120 TD children recording EEG signals at rest using a 10–20 electrode placement system consisting of 8 EEG electrodes (F3, F4, T3, C3, C4, T4, O1, O2). The reference electrode was Cz, and the sampling frequency was 1000 Hz	✓	(Tang et al. 2023)	Accuracy: 87.35%	https://osf.io/uk92c/	12/18/2023
Catarino-EEG (Catarino et al. 2013)	EEG recordings	This dataset contains data on 15 ASD individuals. The average age of these individuals is 31 months, with a standard deviation of 6. Moreover, there are 15 TD subjects with an average age of 29 months and a standard deviation of 4. All the subjects are right-handed males. The diagnosis of ASD for all the subjects is based on the global diagnostic criteria set by clinical psychologists	✓	(Alhassan, Soudani, and Almusallam 2023)	Accuracy: 93.3%, sensitivity: 86.67%, specificity: 100%, Positive PPV: 100%, NPV: 88.2%, F score: 92.86%	https://molecularautism.biomedcentral.com/articles/10.1186/2040-2392-4-1#ref-CR45	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
ASD-Code-and-Datasets (Lakhan et al. 2023)	EEG recordings	The system has analyzed various datasets for three years, including the Ages and Stages Questionnaires (ASQ), Facial Communication and Symbolic Behavior Scales (CSBS), Parents Evaluate Developmental Status (PEDS), Modified Checklist for Autism in Toddlers (M-CHAT), and Screening Tool for Autism in Toddlers and Children (STAT). The system implementation, EdgeX Foundry, and socket programming APIs were used. FCNN, LSTM was designed using Keras and TensorFlow APIs. Additionally, an IoT application was developed using Java and Kotlin	✓	(Lakhan et al. 2023)	Accuracy, Recall, and Precision: 99%	https://github.com/Abdullah-Lakhan/ASD-Code-and-Datasets/tree/main	12/18/2023
William-EEG-DSAD	EEG recordings	The EEG Database for children from 3 to 36 months old tracked the Early Detection of Autism Spectrum Disorder (EEG-DSAD), including 89 low-risk controls and 99 infants with an older sibling diagnosed with ASD	✓	(William J. Bosl, Tager-Flusberg, and Nelson 2018a)	Sensitivity and PPV: 95%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Fan-EEG Data from a Virtual Reality-Based Driving System for ASD Intervention	EEG recordings	Twenty ASD patients (19 males and one female, mean age: 15.29 years) participated in the study. Three subjects had driver's permits, and one had a license. Participants attended six sessions on various days. The Institutional Review Board (IRB) of Vanderbilt University approved using a three-month baseline EEG before and during each session, which lasted 60 min and included three preselected tasks	✓	(Fan et al. 2018)	F1 score: 95%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Sudirman-EEG for Sensory Profile	EEG recordings	The study collected data on sensory responses in autistic children aged 3–7 using unique activities. The senso- ry responses included taste, position, touch, sight, and sound. Data was collected using electrode placement and EEG bio-signals using Neurofax 9200. Of the EEG signals, 64 were recorded, 30 were normal, and 34 were autistic children. The participants were requested to keep their eyes closed during each experiment phase to avoid any distur- bances in the results	✓	(Sudirman et al. 2018)	Accuracy: 99.17%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Shou-EEG Data-Resting-state Gamma	EEG recordings	They included 19 ASD and 19 usually developing patients, matched by age and full-scale IQ. Parents permitted children under 18, and adults gave informed consent. A prior study's subgroup of participants is in this dataset. Sounds attenuated rooms with low light were used to capture EEG data during a 5-min resting state. The EEGLAB toolkit and FASTER plugin preprocessed the data. The cleaned data underwent a down-sampling process to 256 Hz and was then re-referred to the arithmetic expected average	✓	(Shou et al. 2018)		Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Abdulhay-EEG Data-SODP	EEG recordings	The dataset included 120 children, 60 of whom were diagnosed with ASD and the other 60 with typical development. Reputable education experts pre-diagnosed the kids. The selected children were between 4 and 13 years old, and those with additional neurological problems or taking medications were excluded from the study. The selection process was conducted in special education centers and regular schools. The Institutional Review Board ethics committee approved the selection and recording phases, and the parents of the children signed consent forms before participating in the study	✓	(Abdulhay et al. 2018)	Accuracy: 94.4% Sensitivity: 100%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Grossi-EEG Data	EEG recordings	In Villa Santa Maria, Tavermerio (Italy), a certified child and adolescent psychiatrist diagnosed autism in 15 children and adolescents (13 males and two females) using the ADOS scale. The severity score ranged from 4 to 10 points, averaging 7.9. The autistic children were between 7 to 14 years old, with an average age of 10.4 years. None had any hereditary disorders, neuroimaging-documented brain abnormalities, or epilepsy. The socio-economic status, measured by the Hollingshed Index, had an average of 2.35 STD 1.37. The control group had four males and six females aged 7–12 years, with an average age of 9.2 years	✓	(Grossi et al. 2019)	Accuracy: 97.5%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Kang-EEG Data	EEG recordings	The dataset included 97 children: 49 had ASD aged 3–6 years (10 girls, 39 boys), and 48 were typically developing (12 girls, 36 boys). Both groups had similar age and gender distributions. Only typically developing children with no family history of autism were recruited. Experienced psychiatrists diagnosed all subjects. Parents provided consent. Beijing Normal University's ethical committee approved the study based on the Declaration of Helsinki	✓	(Kang et al. 2020)	Accuracy: 85.44%, AUC: 93%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
J. Zhao-EEG Data	EEG recordings	EEG data was collected using an 8-channel setup. Cz was the reference electrode, and electrode impedances were kept below 50 kΩ throughout data collection. The participants were required to sit comfortably on a quiet chair with their eyes open for 5 min while two occipital channels (O1, O2) were analyzed at a sampling rate of 1,000 Hz. These two channels' signals matched the alpha rhythm	✓	(Zhao et al. 2020a, 2020b, 2020c)	Accuracy: 92.66%, AUC: 97.52%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Lekshmylal-EEG Data	EEG recordings	They collected EEG data from 12 subjects aged 4–7 years from SCTIMST in Trivandrum using the Natus machine with a 500 Hz sampling rate. The data was acquired using 10–20 electrode placement international standards, consisting of 18 channels. The left region had odd-numbered connectivity channels, and the right region had even-numbered channels. The silver cup-plated electrodes were used for high accuracy on the central region channels Fz-Cz and Cz-Pz. The data was analyzed using the Natus's machine	✓	(Lekshmylal et al. 2020)		Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Majumder-EEG Data	EEG recordings	A high-density EEG system was used to record the neural signals of 24 children aged 6–13, 12 autistic and 12 typically developing, according to DSM-5 criteria. Subjects were selected based on similar intellectual levels and parental consent. In the experiment, children viewed faces, trees, and cartoons in four blocks. Each block had ten images for each emotion, ten trees, and ten cartoons, repeated twice. For the analysis, only implicit task-related signals were considered, excluding cartoons, to avoid task-related effects	✓	(Majumder et al. 2020)	Accuracy: 90%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Revathi-EEG Data	EEG recordings	EEG signals were recorded from two groups of children with different mental states. The standard group had no mental disorder, while the autistic group had ASD. The children used the neuro-max kit with 16 electrodes to record the EEG signals with 16 channels at 256 Hz. Data was collected using 65 active EEG electrodes. During data acquisition, a syrup named Predictory, containing triclofos sodium, was given to the children to prevent insomnia and flushing	✓	(Revathi et al. 2020)	Accuracy: 90%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Pham-EEG Data-HOS	EEG recordings	EEG signals were collected from a group of children with 37 TD and another group of 40 with ASD between the ages of 4 and 13. The children with ASD were recruited from schools and centers in Jordan after obtaining informed consent from their parents. The criteria for diagnosing ASD were established using well-known tools such as the Childhood Autism Rating Scale (CARS), the Autism Treatment Evaluation Checklist (ATEC), the Psychoeducational Profile (PEP3), and the Social Responsiveness Scale (SRS)	✓	(Pham et al. 2020)	Accuracy: 98.70%, sensitivity: 100%, specificity: 97.30%, PPV: 97.56%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Haputhanthri-EEG Data	EEG recordings	Seventeen participants, with and without ASD, were studied at Indiana University-Purdue University in Columbus, USA. They underwent ADOS-2 EEG with a wireless system (32 channels, 250 Hz) and high-resolution infrared cameras were used to capture thermograms of their faces. The collected data was labeled based on the ASD diagnosis and stored in CSV files	✓	(Haputhanthri et al. 2020)	Accuracy: 94%	Not available online	12/18/2023
Vabalas-EEG Data	EEG recordings	The researchers collected a custom dataset from 44 participants, 22 with ASD and 22 with TD adults, who were matched for handedness, gender, IQ, and age. The dataset contains kinematic and eye movement data recorded while the participants initiated simple hand movements in various conditions	✓	(Vabalas et al. 2020)	Accuracy: 79%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Sundaresan-EEG Data	EEG recordings	Thirteen students (two females) with and without ASD from two schools in California participated in the study. The study used an OpenBCI system, passive noise-cancelling headphones and audio-visual stimuli designed by Muvik Labs. The researchers followed ethical procedures and obtained consent from the participants and their guardians. The STAI-C questionnaire was used to assess trait anxiety in the participants	✓	(Sundaresan et al. 2021)	Accuracy: 93.27%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Baygin-EEG Data	EEG recordings	The collection includes EEG signals from 122 people, including 61 people living with ASD and 61 healthy controls. The participants were selected based on DSMV criteria, neurotypicality, and no co-morbidities or drugs. The recording was done without anesthesia and included easy mimicry. Both groups were matched for handedness, gender, IQ, and age, with ASD encompassing all severity levels. EEG data were split into 6,006 15-s intervals, 4252 from ASD patients and 1754 from healthy controls	✓	(Baygin et al. 2021)	Accuracy: 96.44, sensitivity: 97.79, specificity: 93.16, precision: 97.19 F1-score: 97.49 G_Mean: 95.45	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Yang—EEG Data	EEG recordings	EEG data was gathered from a total of 62 children, comprising 31 with ASD and 31 with TD. The children were shown various movie clips meant to elicit emotional responses. After the data collection, their brain's functional connectivity is analyzed using four commonly used connectivity models: Coherence, Phased Lag Index, Weighted Phased Lag Index, and Phase Locking Value. This analysis was done across five different frequency bands, including theta (4–8 Hz), alpha (8–12 Hz), low beta (12–16 Hz), high beta (16–25 Hz), and gamma (25–45 Hz)	✓	(Yang, Chen, and Cai 2021)		Not available online	12/18/2023
Oh—EEG Data	EEG recordings	EEG signals were collected from 37 typically developing children aged 4–13 years and 40 ASD children with pre-diagnosed behavioral disorders from special education centers and schools in Jordan. Informed consent was obtained from parents before the study	✓	(Oh et al. 2021)	Accuracy: 97.4%, sensitivity: 97.5, PPV: 97.5%, specificity: 97.3%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Alotaibi—EEG Data	EEG recordings	The dataset involved a sample group of 24 children, 12 with ASD and 12 TD. The group comprised seven males and five females, ages 6 to 13, and had an average age of 10.2. Participants were shown three facial expressions during the EEG recordings: joyful, neutral, and scared. The 850 ms blocks displayed ten faces with a 150 ms gap between stimuli. The EEG data was captured at 250 Hz using a 128-channel HydroCel Geodesic Sensor Net	✓	(Alotaibi and Maharatna 2021)	Accuracy: 95.8%, sensitivity: 100%, specificity: 92%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Ranjani—EEG Data	EEG recordings	The dataset contains EEG signals from 38 participants divided into standard, autistic, and epileptic groups. The EEG signals were recorded with a digital amplifier and active electrodes at a sampling rate of 256 Hz through the BC12000 software recording system. The data acquisition system comprises 16 channels, classified according to the 10–20 acquisition system based on international standards. The subjects were categorized based on their age, where 18 typically developing subjects were between 8–15 years old, eight autistic subjects were between 11–17 years old, and 12 epileptic individuals were between 40–60 years old	✓	(Ranjani and Supraja 2021)	Accuracy: 80%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Wadhera—EEG Data	EEG recordings	EEG was recorded for 10 min from 56 participants, including subject 28. The electrodes were arranged using the 10–20 method and included 21 Ag/AgCl electrodes, with two reference electrodes on each side of the mastoid. The data was captured at a sampling frequency of 200 Hz with a 12-bit resolution. The impedance was kept below 5K Ω . The average age of the participants was (± 3.4) years, with a male-to-female ratio of 11:3. Subject 28 had an average age of 13.6 years (± 2.4) and a male-to-female ratio of 5:2	✓	(Wadhera and Kakkar 2021)	Accuracy: 92.34%, sensitivity: 90.87%, specificity: 95.63%, ROC:0.80	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Liao- EEG Data	EEG recordings	In the study, 80 children were examined for ASD; 40 were diagnosed with ASD at an average age of 4.6±9 months, while 40 were TD controls with a mean age of 4.8± 7 months. Tobii Eye Tracker, a video camera, and Emotiv EPOC+ were used to collect data on eye fixation, facial expressions, and EEG signals. The Emotiv EPOC neuroheadset had 14 data acquisitions and two reference electrodes. There were no significant age or gender differences between the two groups. All children underwent a DSM-5 examination to check for any mental or neurological disorders	✓	(Liao, Duan, and Wang 2022a)	Accuracy: 87.5%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
EEG and ET datasets	EEG recordings, Gaze data	They collected EEG and ET data from 90 children aged 3–6, including those with ASD and generally developing youngsters. Signals were recorded for at least 5 min with 128 passive electrodes before ET tests were conducted using an Ag/AgCl array. The Tobii eye tracker recorded gaze behavior at 300 Hz. All subjects were told to view Tiger Qiao Hu1's dramatic visuals. Eye-tracking data was processed using the Tobii fixation filter to examine children's AOI engagement. The Beijing Normal University School of Psychology's Research Ethics Committee authorized this research, and all children gave informed consent with their parents' agreement before enrolling	✓	(Han et al. 2022)	Accuracy: 95.56%, sensitivity: 92.5%, specificity: 98%, AUC(EEG): 0.889, AUC(ET): 0.946, ROC(EEG): 0.965, ROC(ET): 0.984	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Liao- EEG Data	EEG recordings	It contains EEG readings from 10 4- to 7-year-olds. These include eight autistic and two neurotypical youngsters. The EEG data was captured using a 19-channel DABO system by The Minds Lab, Penawar Special Learning Center, Taman Melawati, and Kuala Lumpur. The experiment records brain impulses throughout various tasks. Two times, the individual will open and shut their eyes for 60 s. Watching 60 s of movies to evoke emotions, following a hand movement video, playing a 120-s matching game, and opening and closing their eyes for 60 s will also be done. Autism generally affects fine motor abilities, memory, and attention, which these exams examine	✓	(Zubair, Syed Mashkur, and Nor 2022)		Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Abdulhay—EEG Data	EEG recordings	Collect 120 autistic and neurotypical 4- to 13-year-olds' EEG data using an efficient 64-channel EEG. The technique automatically eliminates motion and myogenic artifacts in noisy situations. Hyperactive or smart kids were excluded from regular lessons. The Institutional Review Board (IRB) of Jordan University of Science and Technology/King Abdullah Hospital granted ethical authorization for the recording. Parents of all participants signed consent paperwork. A small 64-channel device (eego™sports 64, ANT Neuro®) recorded high-density EEG data during sports. Participants sat quietly with open eyes, wearing Waveguard, ANT Neuro® headgear. Recording software evaluated signals, events, and electrode impedance. Sampling was at 500 Hz	✓	(Abdulhay et al. 2023)	Accuracy: 98.9%, sensitivity: 99.2%, specificity: 99.9%,	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Divya—EEG Data	EEG recordings	We were collected from the Sri Ramachandra Medical University (SRMU), Chennai, India. The dataset consists of the EEG signals of 10 children (5 with low-functioning ASD (of 2 females and three males) and 5 with high-functioning ASD (of 2 females and five males)) who performed a cognitive task. The dataset also includes the demographic and behavioral scores of the subjects, such as age, IQ, and CARS-2	✓	(Divya et al. 2023)	Accuracy: 99.9%, sensitivity: 99.8%, specificity: 100%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Ehwzhhq—EEG Data	EEG recordings	The EEG dataset was gathered from various general and special schools in Beijing, China. It comprises 370 multi-channel EEG segments, including 192 EEG segments for 18 ASD children and 178 EEG segments for 16 TD children aged four to seven. The TD children's age range is from 6 to 8 years. Pediatric psychiatrists classified children with ASD at an early stage using the diagnostic criteria for autism outlined in the DSM-5. The EEG data were captured via a 128-electrode array at a sampling rate of 1000 Hz, with each EEG segment lasting 4 s	✓	(Ehwzhhq, Sdwllhqv, and Kh-dowk 2023)	Accuracy: 92%, recall: 96%, F score: 92. AUC: 97%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Manoharan—EEG Data	EEG recordings	The dataset involved 20 children, with 10 having ASD and 10 with TD cognition, between the ages of 5 to 7 years old. The participants' cognitive levels were assessed using DSM V before the study. All experimental procedures were performed at the Sri Ramachandra Medical College Research Institute (SRMC-RI). The participants consented to use their EEG data for research purposes before the study began. The ethical committee at SRMC-RI approved the research, and the study was conducted at Vidya Sudha School, SRMC-RI. Finally, the participants were instructed to sit in a comfortable chair in a quiet, dark room during the study	✓	(Manoharan and Radhakrishnan 2023)	Accuracy: 91.86%, sensitivity: 100%, specificity: 85.6%, MSE	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Wadhera—EEG Data	EEG recordings	The dataset included a sample of 30 children diagnosed with ASD and 30 children who were TD. EEG data were obtained from 21 electrodes placed on the scalp, following the 10–20 international standard. The electroencephalography (EEG) signals were captured for 20 min while the youngsters were subjected to various stimuli, including films and sounds. The dataset also includes the clinical scores of the children with ASD based on the ADOS diagnostic criteria. This dataset was collected in India at the National Institute of Technology, Jalandhar, and the EEG device used was the Nihon Kohden Neurofax MEB9000	✓	(Wadhera, Bedi, and Sharma 2023a)	Accuracy: 97%, AUC: 99.8%	Not available online	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
Visu—EEG Data	EEG recordings	EEG data from healthy people was sampled 256 times following international norms. Five seconds of eye blinks and movements were captured in an annotation file. The time value was preserved after two electrodes were removed in separate directions to mimic motion artifacts. The data was used to test algorithm sensitivity	✓	(Visu et al. 2023)	Accuracy: 97.6%, sensitivity: 94.2%	Not available online	12/18/2023
DANAVA-2 facial image set (Mayor Torres et al. 2022)	facial image	A total of 88 participants (15.34 ± 1.58 years) participated. Among the participants, 48 (29 male, 15.39 ± 1.55 years) had TD diagnosis, whereas 40 (32 male, 14.77 ± 2.16 years) had ASD. ASD patients were diagnosed using the Autism Diagnostic Observation Schedule, Version 2.0 (ADOS-2) and deemed high-functioning on the Kaufman Brief Intelligence Assessment-2. ASD had greater ADOS-CS than TD ($p = 0.038$), but intellectual performance was similar ($p = 0.227$)	✓	(Torres et al. 2023)	Accuracy: 93.4% precision: 93.5%, recall: 93.3%, F1 score: 93.4%	https://github.com/albermax/investigate	12/18/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
SEED	EEG recordings	The dataset consists of information regarding the emotional classification of 15 participants. The data was collected using 62 EEG electrodes, and emotions were categorized as positive, negative, or neutral based on the valence scale used. The dataset includes emotional classification for 15 stimuli and is equally represented across all three emotion categories. Also, the dataset is gender-balanced	✓	(Aslam et al. 2022)	Accuracy: 98%	SEED Dataset (sjtu.edu.cn)	12/19/2023
DREAMER (Katsigiannis and Ramzan 2018)		Data collected from 14 EEG electrodes created EEG and ECG records for 23 individuals. The provided valence, arousal, and dominance categories can help identify and categorize emotions. However, it is essential to note that the sample imbalanced gender and valence classes. Valence and arousal labels can be found on a scale of 1 to 5	✓	(Aslam et al. 2022)	Accuracy: 98.6%	https://github.com/dream2020/data	12/19/2023

Table 1 (continued)

Dataset	Type of Data	Characteristics	Dataset Primary Secondary	Techniques	Implementation	Available online	Accessed
ODU		The dataset includes EEG data from 17 participants, including 8 individuals with ASD and nine individuals with TD. The EEG data was recorded using 32 electrodes, but the signals were only recorded for 14 electrodes across all participants. Additionally, the dataset comprises the ADOS-2 scores of each participant. The dataset consists of 10 males and seven females	✓	(Aslam et al. 2022)	Accuracy: 95.5%	Not available online	12/19/2023

The sample size of available datasets for the diagnosis of ASD may vary depending on the number of patients involved. Most datasets are generated through multiple examination experiments, while a few are collected from other available datasets considered secondary data (i.e., after analyzing the primary dataset). However, it is worth noting that secondary datasets have significantly contributed to the diagnosis of ASD, too (Fig. 16).

Table 2 presents the most used datasets for the diagnosis of ASD, considering sample size, age range, gender distribution, and performance metrics such as accuracy, sensitivity, and specificity. Numerous studies have used these datasets to test the efficacy of ML and DL models. The sample size and the number of studies for each dataset are compared in Fig. 17 to present how each dataset has been used in visual research.

3.3 EEG pre-processing techniques

EEG pre-processing refers to a series of signal-processing techniques applied to raw EEG data. It involves eliminating noise and artifacts, improving signal quality, and extracting relevant features for further analysis and interpretation. The process offers several benefits, including increased signal-to-noise ratio (SNR). That means the brain activity of interest is more distinguishable from background noise and interference, resulting in more accurate and reliable results in EEG-based applications like diagnosis, classification, or brain-computer interface (BCI). Additionally, EEG pre-processing reduces the dimensionality and complexity of the EEG data, making it easier to apply advanced methods and algorithms like machine learning and deep learning (Choi and Kim 2018).

Filters enhance data quality and minimize noise in time–frequency (TF) analysis. They are employed at different phases of the measurement-to-publication process, such as in the measuring rig or amplifier, the analog-to-digital converter, preprocessing, and data analysis. Filters are widely used in measuring and studying brain data, including high-pass, low-pass, notch, band-pass, and time–frequency analysis.

Usually, filters in recording rigs use analog filters in the initial stages (shown in green) and digital filters in subsequent stages (shown in red). High-pass filters remove slow drift components, while low-pass filters are used to attenuate noise or prevent antialiasing. Data analysis often involves band-pass filtering or TF analysis, while data display or plotting may require additional smoothing. Figure 18 illustrates where the applied filters are used.

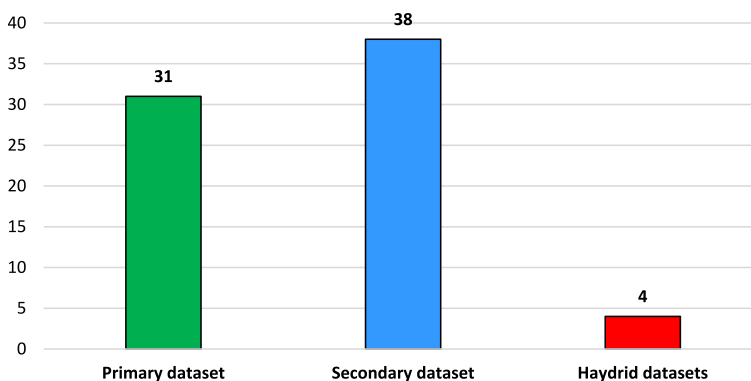


Fig. 16 Statistics publications in Autism Spectrum Disorder

Table 2 Comparison of the most commonly used datasets in ASD diagnosis

Dataset	Sam- ple Size	Age Range	ASD Gender Distribution (M/F)	Number of Studies	Best Per- formance Metrics	Avail- able Online
KAU	18	9–16 years	8 M/10 M	10	Accuracy: 99.9%, Sensi- tivity: 100%, Specificity: 97%	Yes
ADOS-2-ASD-EEG	8 ASD, 9 TD	5–16 years	6 M / 2F	4	Accuracy: 90%, Sensi- tivity: 93%, Specificity: 92.4%	No
BCIAUT-P300	15 ASD	Various	Not Specified	7	Accu- racy: 93.16%, Sensitiv- ity: 94.65%, Specificity: 91.4%	Yes
VSM-EEG	15 ASD, 10 TD	7–14 years	Not Specified	3	Accu- racy: 100%, Sensitiv- ity: 94.95%, Specificity: 100%	No
SAFARI	41 ASD, 16 TD	10 months—21 years	Not Specified	2	Accuracy: 92.2%, Sensi- tivity: 94.5%, Specificity: 90.5%	No
ASD-EEG	60 (30 ASD, 30 TD)	4–8 years	Not Specified	5	Accu- racy: 81.96%, Sensitiv- ity: 66.66%, Specificity: 94.11%	No
DEAP	32	14–16 years	Not Specified	2	Accuracy: 63%	Yes

M stands for male, F for female, and TD means typically developing person who does not have autism

3.3.1 Low pass filter and high pass filter

Two types of filters can remove unwanted noise or artifacts from EEG signals. The low-pass filter allows signals below a particular frequency, called the cutoff frequency, to pass through while weakening signals above it. On the other hand, the high-pass filter allows signals above the cutoff frequency to pass while reducing signals below it. However, it is essential to remember that these filters may introduce distortions or delays in the EEG signals. Therefore, considering the research question and experimental design, their use should be carefully considered (Newman 2023). The diagram in Fig. 19 shows the low pass filter in red, the bandpass filter in blue, and the high pass filter in green for signal transmission.

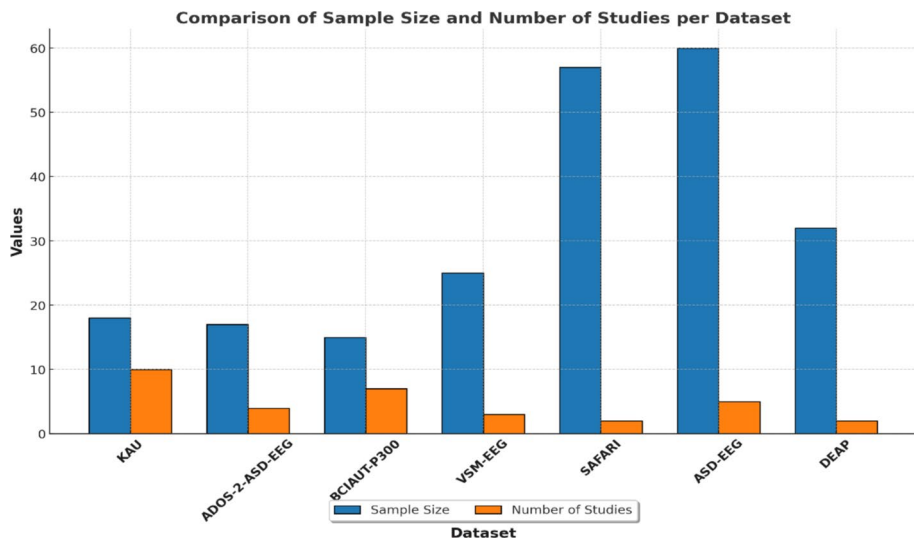


Fig. 17 Bar chart comparing sample sizes and the number of studies for each dataset

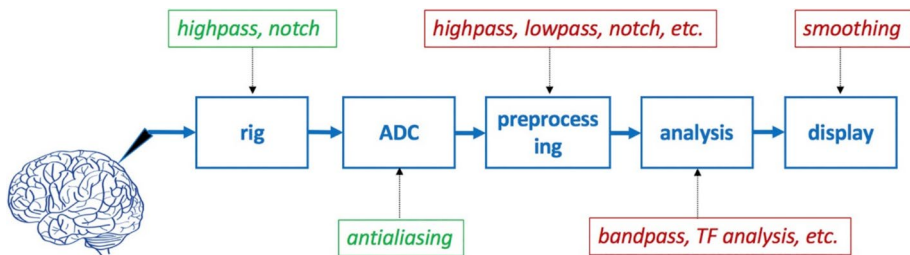


Fig. 18 Applied Filters Areas (Cheveigne 2019)

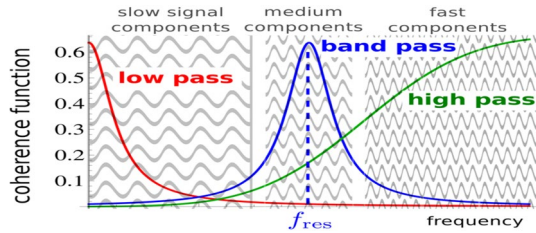
3.3.1.1 Low-pass filter A low-pass filter with a 40 or 50 Hz cutoff frequency eliminates high-frequency noise from EEG data, such as EMG activity or radio frequency interference (Manoharan and Radhakrishnan 2023).

3.3.1.2 High-pass filter A high-pass filter with a 0.1 or 1 Hz cutoff frequency can eliminate low-frequency drifts or baseline changes from EEG data, such as EOG activity or electrode polarisation (Divya et al. 2023).

More than 25 studies utilized both low-pass and high-pass filters, with the best outcomes being:

(Bakheet and Maharatna 2021) An EEG dataset from 24 subjects examined the effects of neutral, happy, and fearful facial expressions. The data underwent filtering using 0.5 Hz for high-pass and 50 Hz for low-pass filters and was segmented into 1000 ms timeframes. Any signals above 200 μV were removed as artifacts. The data was then subjected to band-pass filtering and baseline adjustment to achieve 100% accuracy in the happy stimulus dataset.

Fig. 19 Three filter characteristics (Schreiber, Lindner and Blankenburg 2016)



(Manoharan and Radhakrishnan 2023) used several techniques to remove 50 Hz power line interference noise from EEG signals. They applied low-pass, high-pass, and notch filters and used the wavelet denoising method to preserve signal characteristics and remove noise. The *coif5* wavelet with SURE soft thresholding based on MSE was the most effective among the various algorithms tested. The researchers segmented the denoised signals into 120-s epochs and achieved an accuracy of 91.86%.

3.3.2 Band pass filter

This filter allows signals within a specific frequency range, the passband, to pass through while attenuating all signals outside this range. It eliminates unwanted noise or artifacts from EEG signals, such as low-frequency drifts, high-frequency muscle activity, or power line interference. A bandpass filter can also isolate a specific frequency band of interest, such as alpha, beta, delta, or theta waves, from the EEG signals (Newman 2023). Figure 20 shows an example of using a bass band filter.

There are various methods to apply a band-pass filter to EEG signals, which can vary based on the type of filter, cutoff frequencies, and the software or tool utilized. Some commonly used filters include Butterworth, Chebyshev, Elliptic, and Bessel filters, which differ in their steepness, ripple, and phase distortion characteristics and trade-offs (Kurapa et al. 2020).

This section focuses on the often-employed pre-processing methods represented by the Band Pass Filter used in more than 35 studies and achieved results of more than 98% such as:

(Ibrahim et al. 2018) They used the KAU dataset, which comprised sixteen subjects aged between 9 and 16. Of the sixteen, ten were normal, while nine had autism. The dataset was acquired using a 16-channel EEG with a sampling rate of 256 Hz and gathered using the G.Tech software and hardware package. The data underwent preprocessing using an elliptic bandpass filter, limiting the signals to the frequency range of 0.5 to 60 Hz and leading to a remarkable accuracy of 99.71%.

Also, (Alturki et al. 2020) participated in the pre-processing phase, using the Independent Component Analysis (ICA) approach and adaptive filtering to eliminate ocular artifacts. The KUA dataset is divided into equal segments and then subjected to a filtering process to eliminate any noise and interference. The elliptic bandpass filter (0.1–60 Hz) offers superior experimental outcomes to other filters due to its efficient memory and computation requirements, reducing delay time. The result yielded a level of accuracy of 99.9%.

Another study used a bandpass filter with a pass frequency band (0.1–60 Hz) with an accuracy of 99.15%. Moreover, (Divya et al. 2023) used a Butterworth bandpass filter to extract signals in 0.5–40 Hz with an accuracy of 99.9%. (WorldPopulationReview 2023).

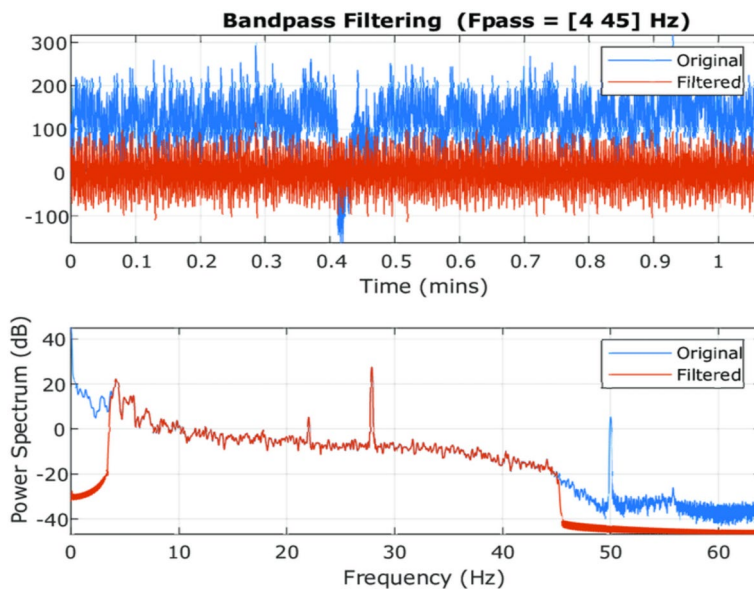


Fig. 20 Individual EEG channels with power spectra band-pass-filter (Dar, et al. 2020)

3.3.3 Notch filter

A notch filter is a type of filter that allows only those signals that lie outside a specific frequency range to pass through, known as the stopband. The filter suppresses or attenuates all signals that fall within this range. This filter is beneficial for removing specific frequencies, such as power line noise and other physiological signal artifacts, from EEG signals while preserving the desired EEG information (William O Tatum, Anteneh M Feyissa 2023). The notch filter is shown in shown in Fig. 21.

Studies used a notch filter for 11 studies. The highest results were when using a notch and bandpass filters with the KAU dataset. The best results were:

Tawhid et al. (2021b) The KAU dataset was filtered using a bandpass and notch filter, and all EEG signals were digitized at a 256 Hz sampling rate. The proposed method pre-processed and normalized the data using CAR and IIR, leading to an accuracy of 99.15%.

(Thirumal and Thangakumar 2022) Utilizes EEG and behavioral information from the KAU BCI Group to screen toddlers for autism. The EEG data comprises ten traits that measure autistic characteristics. The dataset evaluates the efficacy of feature selection techniques and classifiers in diagnosing ASD. The EEG data undergoes filtration via a bandpass filter and a notch filter. The data is then converted into digital form at a sampling rate of 256 Hz. The investigation yielded a precision rate of 97.15%.

3.3.4 Other methods of pre-processing

Other studies have used different pretreatment methods, such as:

(Kumar et al. 2019) the signals were preprocessed using a Kalman filter. (Aslam and Altaf 2019) The DEAP dataset already pre-processes the EEG data. The present work

(Bouallegue et al. 2020) uses fundamental FIR/IIR techniques and advanced methods to develop a comprehensive filter design. Six filters are customized for each channel's epoch level of the EEG data to generate the filtered signal. This filtered signal serves as the input for training an RNN with GRUs. Figure 22 displays an example of a response obtained using a time-varying Kalman filter.

A study (Han et al. 2022) was conducted on 90 children aged 3–6 years, including 40 with ASD and 50 typically developing children. The researchers used various questionnaires to assess the participants, recorded EEG data, and processed eye-tracking data using the Tobii TX300 eye tracker. The EEG data was preprocessed using downsampling, notch filtering, and bandpass filtering techniques. Eye-tracking data was also pre-processed to detect screen fixations and replace missing gaze data.

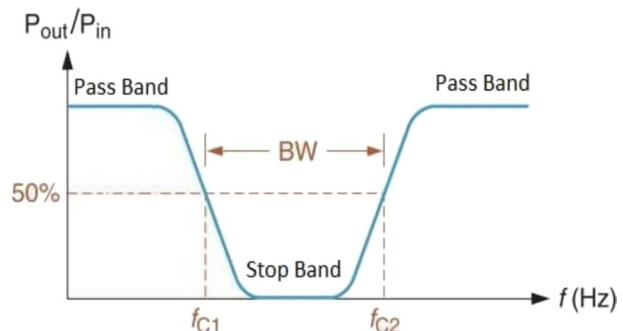
Eight works used EEGLab in MATLAB for data pre-processing, such as performed data pre-processing via Matlab R2016a and EEGLab. The maximum classification accuracy achieved was 85.44%, with an AUC of 0.93, using 32 selected features. At the same time, the best results were by (Wadhera, Bedi, and Sharma 2023b), Where raw EEG signals are imported to MATLAB and preprocessed using EEGLAB. Refers to several processes, including independent component analysis (ICA), channel re-referencing, noise reduction, removing artifacts, and interpolation removal. The suggested model outperformed state-of-the-art approaches with 97% accuracy and 0.998 AUC.

Table 3 shows the results achieved by various filters. Evaluation metrics indicate an accuracy of 90% and higher.

3.4 EEG features extraction techniques

EEG feature extraction involves extracting relevant and discriminative information from EEG signals. It has various applications such as diagnosis, classification, prediction, and intervention of brain-related disorders or functions. Multiple domains, including time, frequency, spatial, time–frequency, and decomposition, are available for EEG feature extraction. Features may be extracted from EEG data using a variety of methodologies and procedures, each with its own set of benefits and drawbacks. Transforming the raw signal into a feature vector, a pertinent data structure that removes noise and emphasizes important data, happens during the feature extraction step. To streamline the classification process, it could also include “dimensionality reduction” to remove unnecessary data from the feature vector (Amin et al. 2015, 2017; Hu and Zhang 2019).

Fig. 21 Notch Filter (Rajiv 2022)



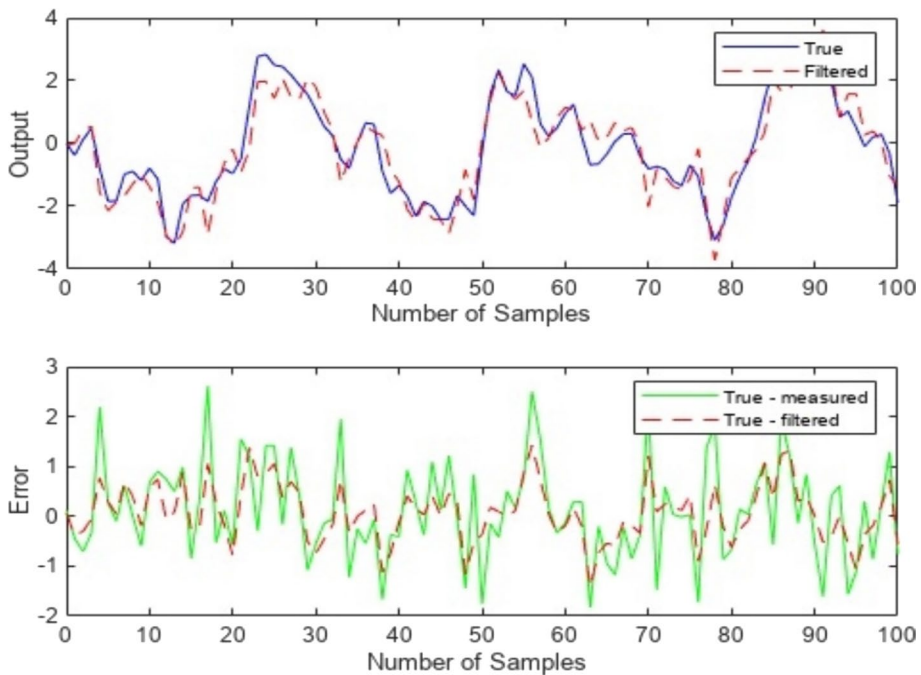


Fig. 22 Response with Time-Varying Kalman Filter (Athans 2018)

EEG feature extraction is used for various applications, including diagnosing neurological disorders, brain-computer interfaces, and cognitive neuroscience. There are many methods for EEG feature extraction, depending on the domain, type, and purpose of the analysis. Some common domains include:

1. **Time-domain features (TDF):** Stationary signals are the realm of time-domain analysis, which does not apply to biosignals. Quantifying a non-stationary time series is possible by considering several stationary parts. Their temporal domain features indicate the amplitude, duration, and shape of EEG signals. There are three groups, each consisting of twelve notable traits. (1) the average and standard deviation for a normally distributed set of data; (2) the center, mode, range, first and third quartiles for a geographical assessment; (3) the maximum and minimum variation, skewness, and kurtosis for a form determination. Some more methods for obtaining EEG parameters include the slope-signal change, Willison amplitude, Lyapunov exponents, and Hjorth parameters. Without transformation, they are quick and easy to calculate (Kim et al. 2020; ALYASSERI 2020; Singh and Krishnan 2023a, 2023b).
2. **Frequency-domain features (FDF):** Examples of frequency-domain analysis methods include autoregressive, Eigenvector, and Fast Fourier Transform (FFT). The Short-Time Fourier Transform (STFT) is preferred over the Fast Fourier Transform (FFT) in biosignal analysis because it separates a function into its frequency components. Eigenvectors are used to compute signal frequency and power from observations that have artifacts, resulting in high-resolution frequency spectra. Pisarenko, MUSIC, and

Table 3 Pre-processing for EEG signals using various techniques

#	Authors	Description of pre-processing steps
1	William J. Bosl et al. (2018b)	The signals were processed by amplifying them, applying a bandpass filter ranging from 0.1 to 100.0 Hz, and then sampling at 250 Hz or 500 Hz. The signals sampled at 500 Hz were later downsampled to 250 Hz for further analysis. A 12-bit National Instruments Board (National et al.) digitized the signals
2	Fan et al. (2018)	Include slew rate limiting, bandpass filtering, segmentation, artifact removal, and correction
3	Ibrahim et al. (2018)	bandpass filtering (an elliptic bandpass filter)
4	Abdulhay et al. (2018)	A bandpass filter with a range of 0.3–40 Hz was applied to the recordings, and the LA-106 ASA ERP software was used to remove EEG signal artifacts
5	Jayawardana, Jaime, and Jayarathna (2019)	The EEG data in its raw form was brought into EEGLAB, which is a MATLAB toolbox designed explicitly for EEG analysis. It was then filtered using a band-pass filter, ranging from 0.1 Hz to 100 Hz, and a notch filter of 50 Hz to eliminate low-frequency drifts and power-line noise. The data was then divided into four epochs, namely, BASELINE, START, MIDDLE, and END, and re-referenced to the average of all electrodes to reduce the effect of reference electrode noise. The data was cleaned using ICA to remove non-brain sources of variance, such as eye blinks and muscle artifacts. Furthermore, each channel subtracted the mean to normalize the data and divided by its standard deviation
6	Peya et al. (2020)	The EEG signals were recorded using a Micromed device that had SystemPlus Evolution software. The electrodes on the scalp were located using the International 10–20 System bandpass filter, which captured only the frequency components between 0.3 and 70 Hz
7	Alturki et al. (2020)	During the pre-processing stage of analyzing EEG data, the first step is to use Independent Component Analysis (ICA) and adaptive filtering techniques to remove eye artifacts. The EEG dataset is then divided into segments of equal length to ensure that each segment contains the same information. Once the EEG signals have been segmented, they are filtered to eliminate any noise or interference that may have been generated during the recording process. An Elliptic bandpass filter with a range of 0.1–60 Hz is the best choice for this task since it requires less memory and calculation time and provides a shorter delay time than other types of filters
8	Majumder et al. (2020)	The data was filtered using a bandpass filter ranging from 0.01 to 40 Hz. It was then averaged and referenced. During the trials, ocular artifacts such as blinks and eye movements were identified using bipolar EOG channels and visual examination. These artifacts were then removed from the study. The EEG signals were synchronized with the start of the stimulus and consisted of a 150 ms period before the stimulus and an 850 ms period after the stimulus
9	Revathi et al. (2020)	EEG data needs pre-processing to mitigate artifacts caused by movement, environmental factors, and electrodes. Channels are adjusted by subtracting the average value and then filtered with a bandpass filter of 0.1 to 60 Hz. Finally, data is normalized to 0–1
10	Pham et al. (2020)	This research captured 64-channel brain waves at 500 Hz for 20 min. These signals were band-pass-filtered to encompass 0.3–40 Hz. Signals were filtered and divided into 5519 samples for investigation. Extracting nonlinear characteristics from higher-order spectra (HOS) bispectrum displays the required math. These characteristics were analyzed statistically
11	Rechal, Kumar, and Khaleelulla (2021)	The VMD technique processes signals that break down EEG signals into limited modes with various spectral bands and center frequencies. It helps lower noise, reveal features, and make complicated signals easier to understand
12	Mohi-Ud-Din and Jayanthi (2021a)	The data was processed using a bandpass filter, removing frequencies outside the range of 0.5–50 Hz of interest or noise and excluding data from channels 125, 126, 127, and 128, which were found to be noisy and unreliable

Table 3 (continued)

#	Authors	Description of pre-processing steps
13	Tawhid et al. (2021a)	During recording, the datasets were filtered using a band-pass filter with a passband of 0.1–60 Hz and a notch filter with a band of 60 Hz. Finally, all EEG signals were digitized using a 256 Hz sampling rate. The recommended approach starts with data pre-processing, which involves normalization, common average referencing (CAR), and infinite impulse response (IIR) filtering
14	(Alotaibi and Maharatna 2021)	There are several steps involved in analyzing EEG data. In the first step, the data is divided into 1000 ms epochs. In the next step, any artifact-ridden epochs are removed, and a frequency band filter (alpha, beta, delta, theta, and gamma) is applied. Finally, the phase-locking value (PLV) matrix is computed
15	Alturki et al. (2021)	The EEG datasets underwent ICA processing to remove artifacts and were then filtered with a fifth-order band-pass Butterworth filter to eliminate noise and interference
16	Bakheet and Maharatna (2021)	EEG data was separated into 1000 ms chunks to focus on the event window. Epochs with signals exceeding 200 millivolts were rejected as artifacts. A fifth-order forward-backwards Butterworth filter with cut-off frequencies from 0.5 to 50 Hz was used to eliminate drift, low frequency, and noise from measurements. After a baseline adjustment, eight Fusiform Gyrus channels were chosen
17	(Wadhera and Kakkar 2021)	Muscle contamination and artifacts were manually removed from the data after applying bandpass filtering within the 0.1–40 Hz range. All samples exceeding $\pm 100 \mu\text{V}$ were eliminated. The data was further processed using EEGLAB's ICA to eliminate visual artifacts. The data was extracted into continuous 5.12-s segments of 1024 samples per participant and channel to ensure efficient processing and better quality
18	Wadhera and Mahmud (2022)	A bandpass filter is applied to process an EEG signal to extract the main frequency band (0.1–40 Hz). Artifacts are then removed by visually inspecting the signal and discarding any physiological noise, eyeblinks, or large amplitudes. The runica() function from the EEGLAB toolbox further eliminates artifacts. Next, the EEG data is converted into node and edge forms to create GCN (Graph Convolutional and Network) networks
19	Han et al. (2022)	The EEG data was pre-processed using MATLAB and EEGLAB. The pre-processing steps included • downsampling the data to 250 Hz to reduce its size and complexity • A 50 Hz notch filter was applied to remove line noise • A band-pass filter (0.5–45) eliminates high-frequency and low-frequency noise Eye-tracking (ET) data is preprocessed to detect screen fixations and replace missing gaze data using linear interpolation for up to 100 ms. Participants who spent over 70% of their time staring at screens provided the data. The EEG data was separated into four non-overlapping segments to remove distortions such as eye movements and outlier values. Visual assessment rejected segments with noise using a $200 \mu\text{V}$ threshold to identify and interpolate the wrong channels. So that the research could cover as much of the brain as possible, 62 electrodes were chosen from the 128-channel GSN
20	Alhassan and Soudani (2022)	Data was gathered from 32 channels using a 10–20 system. A reference electrode was positioned at the tip of the nose. The data was recorded at a sampling rate of 1000 Hz and filtered with a bandpass filter with a 0.1–50 Hz range
21	Thirumal and Thangakumar (2022)	The preprocessing includes applying a band-pass filter (0.1–60 Hz) and a notch filter (60 Hz), and then the EEG data is digitized at a frequency sampling rate of 256 Hz
22	Divya et al. (2023)	The recorded signals have been adjusted using the average reference to ensure uniformity. The initial data was pre-processed using Independent Component Analysis (ICA) in EEGLAB to eliminate motion artifacts. A Butterworth bandpass filter was used to isolate signals within the 0.5 to 40 Hz frequency range, and the min–max methodology was used to normalize the data. The Savitzky-Golay (SG) filtering method was used to smooth the data

Table 3 (continued)

#	Authors	Description of pre-processing steps
23	Wadhera, Bedi, and Sharma, (2023a)	To process the data in Dataset I, they used a bandpass filter with a frequency range of 0.1 to 60 Hz and a notch filter at 60 Hz to reduce unwanted noise and artifacts from Dataset I during the data processing stage. In addition, we recorded the EEG signals using a sample frequency of 256 Hz. Finally, we divided the EEG data into 20-min intervals for each participant The raw EEG signals are imported into the MATLAB toolbox to process Dataset II. Then, EEGLAB will be used for pre-processing, which involves ICA, channel re-reference, noise reduction, artifact removal, and interpolation removal. Once the pre-processing is complete, we will partition the EEG signals into four distinct epochs: baseline, start, middle, and end epochs. Every epoch will include a particular sample size In Dataset III, the EEG data will be re-referenced using the average mastoid signal with fewer brain signals. The runica function in the EEGLAB toolbox will eliminate artifacts from the EEG data. The EEG data will undergo visual inspection to eliminate physiological noise, instrument drift, eye blinks, and large amplitudes. Ultimately, the EEG signals will be divided into non-overlapping epochs of 4 s each
24	Ehwzhq, Sdwlhqvw, and Khdownk (2023)	Electrode localization was used to select a total of 19 EEG channels. The EEG data was then filtered using a band-pass filter with a 0.3–80 Hz range. Spherical splines were used to interpolate deficient activation channels. Furthermore, independent component analysis (ICA) was used to separate the multi-channel EEG data, which helped identify spatially fixed and temporally independent components, which in turn helped remove eye movement and muscle activity. The average reference method was used for re-referencing
25	Alhassan, Soudani, and Almusallam (2023)	19 EEG channels were selected using electrode localization and filtered using a 0.3–80 Hz band-pass filter. Spherical splines were used to interpolate deficient channels. Independent component analysis (ICA) removed eye movement and muscle activity. Finally, the average reference method was employed for re-referencing
26	Mohi Ud Din and Jayanthi (2023)	The EEG data processing involved several steps. Firstly, EEGLAB's FIR filter was used to apply a band-pass filter ranging from 0.5 to 50 Hz. Secondly, channels 125 to 128 were removed as they were not present in the original dataset. Thirdly, EEG fragments of 8 s were extracted from each individual. Finally, the ASD signals were labeled as "ASD," while the standard signals were labeled as "NOR."
27	Sivasaravana Babu et al. (2023)	Correct baseline shift in EEG signals by subtracting the pre-event mean amplitude from the signal amplitude. Divide the signals into five frequency subbands and perform a discrete wavelet transform (DWT) using a Daubechies wavelet of order four (db4) and five decomposition levels. The DWT approximates and details coefficients for each decomposition level, reflecting the EEG signal's low- and high-frequency components. Each level yields one approximation coefficient and two detail coefficients
28	Sudirman et al. (2018)	The pre-processing step of the EEG signal comprises the analysis of removing artifacts and reducing noise using Independent Component Analysis (ICA) with the EEGLAB program on the collected raw EEG data
29	Heunis et al. (2018)	The EEG data processing involved several steps. Firstly, lengthy raw EEG segments were clipped, and artifact-containing epochs were removed. The data was then resampled to 200 Hz and notch-filtered at 60 Hz. Spatial down-sampling was performed on 19 electrodes, excluding the Fp1 and Fp2 channels. An average reference was created, and FIR filtering was applied from 1 to 70 Hz. Artifact-filled periods were manually discarded, and awake-task-free data was selected. Muscle-contaminated epochs were removed, ocular artifacts were fixed with independent component analysis, and 5-s segments were taken out continuously for each person
30	Alturki et al. (2019)	EEG data is segmented into specific lengths, windowed to reduce spectral leakage and filtered to remove unwanted artifacts from 0 to 60 Hz low-pass elliptic filter
31	Haputhanthri et al. (2019a)	The raw EEG signal is scanned to identify values outside a predefined threshold T0. These outliers are then replaced with the cumulative mean, calculated as the average value of the previous data points. Furthermore, any noise caused by eye-blink is filtered out, and the signal is smoothed

Table 3 (continued)

#	Authors	Description of pre-processing steps
32	Kumar et al. (2019)	The signals are first preprocessed using a Kalman filter (KF). This filter is an efficient computational tool that helps evaluate a method's state by reducing errors. Next, the signal is decomposed using variable mode decomposition (VMD)
33	Grossi et al. (2019)	The MS-ROM/IFAST method is a preprocessing technique to convert EEG data into 1-dimensional vectors. It employs unsupervised neural networks, multiscale entropy methodology, and self-organizing maps to project the vectors onto a 2-dimensional grid. The grid cells are then ranked based on frequency, resulting in a topological vector. The technique eliminates noise and selects relevant features based on the correlation matrix. The objective is to detect consistent patterns in the spatial distribution of EEG across time and prepare the data for classification
34	Bouallegue et al. (2020b)	The primary aim of this study is to preprocess EEG data using fundamental FIR and IIR filters and cutting-edge approaches. Six filters are carefully chosen and placed at the epoch level to make a better signal. This signal then trains a complex recurrent neural network (RNN) with gated recurrent units (GRUs). The suggested recurrent neural network (RNN) uses a specific filter with a customized order to minimize its mean squared error (MSE) at the epoch level. The suggested method uses a dynamic filtering method to improve classification accuracy. It combines independent component analysis (ICA) with a convolutional neural network (CNN) for the best decoding performance
35	Zhao et al. (2020a, 2020b, 2020c)	The EEGLAB toolbox, a MATLAB-based software, analyses EEG data by eliminating noise and detecting adaptive artifacts. The data is divided into non-overlapping 4-s time series, and interfering signals like EMG and breathing are removed. The data is then sampled to 200 Hz for accuracy. If the artifact component exceeds a set threshold, the period is discarded
36	Grossi et al. (2020)	The study employed a filtering process to choose variables based on their correlation with the diagnostic class. That resulted in fewer variables, with 100 selected for study 1 and 72 for study 2. The dataset underwent filtering and was then divided into two sets, one for training and one for testing. The TWIST method was used for partitioning. The training set had 12 extracted features, while the testing set had 36. The data was analyzed through Sine-net and Back Propagation ANNs, utilizing a training–testing strategy on the preprocessed datasets from TWIST
37	Zhao et al. (2020a, 2020b, 2020c)	The pre-processing of raw EEG data involves several steps that aim to reduce noise, extract essential features, and enhance the performance of classifiers. These steps include feature extraction, low-pass filtering, non-target ERP (NT-ERP) based filtering, and feature selection. The article comprehensively elucidates the objectives and methods of every stage in the pre-processing pipeline
38	Haputhanthri et al. (2020)	The EEG pre-processing pipeline comprises four phases to enhance data quality and precision. The first is channel re-referencing, which converts EEG signals into a common reference point. The second step removes low-frequency baseline activity, the third removes AC powerline noise, and the final step removes eye-blink artifacts using techniques like ICA and template matching
39	Baygin et al. (2021)	The 1D_LBP technique extracts nine-dimensional microstructures from the EEG segment and creates a feature vector with a dimensionality of 16,384. The feature vector undergoes a short-time Fourier transform to form spectrogram images. This process breaks down the signal into spectra, each corresponding to a short-time segment. These graphics display the frequency and amplitude changes of the signal over time and act as representative examples for each class
40	Dong et al. (2021)	The research filtered, cleaned, and reduced the sampling rate of EEG data using the EEGLAB toolbox. The data was reduced to 500 Hz to enhance its noise resistance. The data was transformed from the time domain to the frequency domain using DFT and a hamming window for feature extraction. The study used a personal computer with an Intel (R) i5-8400 CPU, 16 GB of RAM, and Windows 10

Table 3 (continued)

#	Authors	Description of pre-processing steps
41	Oh et al. (2021)	The whole dataset consisted of 5519 data points. The research used an EEG cap to capture 20 min of brain waves from subjects' scalps. The cap was linked to a 64-EEGO amplifier and a desktop computer. The signal was recorded at a sampling rate of 500 Hz, covering a frequency range (0.3–40) Hz
42	Ari et al. (2022)	The pre-processing steps include using the Douglas-Peucker algorithm to reduce input EEG signal samples, extracting EEG rhythms using wavelet transform, coding them using sparse representation, converting them into decimal numbers with 8-bit lengths, and concatenating the histograms to form an image by concatenating the decimated rhythm signals
43	Aslam et al. (2022)	The study analyzed EEG data using an adaptive artifact detection technique and manual identification, eliminating interference and artifacts. Channel selection techniques were used to determine the optimal subset of channels, followed by feature extraction and feature selection algorithms to compute 32 features
44	Abdulhay et al. (2023)	All collected EEG signals have been manually reviewed and corrected for artifacts, noise, and muscle-related activity using the ANT Neuro® software eego™sports and ASA 4.10.1. A neurologist has also verified the data by visually inspecting them and then using a semi-automatic tool to validate them
45	Tawhid et al. (2023)	The EEG signals underwent resampling to a standardized frequency of 256 Hz and were reduced to 16 channels across all datasets. They were then re-referenced using the common average referencing (CAR) technique to minimize noise and enhance spatial resolution. To eliminate artifacts from muscle activity, ocular noise, and external interference, a low-pass IIR filter at 40 Hz is applied. Finally, the signals were normalized using the Zero Mean Unit Variance (ZMUV) method to reduce variations between individual signals and simplify computational processes
46	Chen, Gaoxiang, and Li, (2023)	The EEG data was first resampled to 250 Hz for uniformity. Divided into 4 s continuous data epochs, artifacts and noise were removed, and electrode locations were converted to unified spherical coordination using EEG lab. Data was extracted from 32 channels and processed for each channel using continuous wavelet transform (CWT), and temporal and spectral dimensions were resampled to 32 for consistency. Finally, the data was stacked according to the channel sequence to create a temporal-spectral-spatial representation of the EEG epoch
47	Visu et al. (2023)	The study introduces a new technique called Adaptive DWT, which combines DWT's temporal resolution with Fourier transforms to remove signal artifacts. This technique uses adaptive thresholding to overcome premature wavelet-based filtering convergence and, thus, reduce iterations. The ICA decomposes the signal, separating its components from the mixed signals. The study records eye blinks and movements for five seconds, simulating motion in individuals. Moreover, data was analyzed by EEGLAB with a focus on algorithm performance and sensitivity
48	Manoharan and Radhakrishnan (2023)	A notch filter eliminates 50 Hz power line interference noise from raw EEG data, filtered using low-pass and high-pass filters with cut-off frequencies ranging from 0.53 to 70 Hz. After considering several algorithms, the <i>coif5</i> wavelet with SURE soft thresholding emerged as the top choice according to the mean square error (MSE) criteria. Its purpose was to preserve the signal's characteristics while filtering background noise. To better match the denoised signals with the stimuli shown to youngsters, we divided them into 120-s epochs
49	Mayor Torres et al. (2023)	The EEG data was first filtered at 60 Hz, down-sampled to 500 Hz, re-referenced, and then segmented into epochs ranging from 0 to 1500 ms. Machine learning approaches such as Makoto's pipeline for incorrect channel elimination and Koethe's clean raw Artefact Subspace Reconstruction were used to clean each trail automatically. Using the ADJUST EEGlab plugin, two EEG channels had their blinking and movement artifacts eliminated. After eliminating artifacts in the EEG, the trials were adjusted for baseline using linear de-trending

Minimum-Norm are some high-resolution eigenvector algorithms. Autoregressive methods estimate EEG power spectrum density PSDs parametrically, reducing spectral leakage and improving frequency resolution. Burg uses the Levinson-Durbin technique to estimate reflection coefficients and parameters. Frequency domain properties like FFT, PSD, band power, and spectral entropy can detect signal periodicity, oscillations, and energy distribution. These properties take longer to compute than time-domain features (Singh and Krishnan 2023a, 2023b; ALYASSERI 2020; Kim et al. 2020).

3. **Time and frequency features (discretely):** Classification accuracy might be enhanced by merging time- and frequency-domain data. Using time-domain features Discrete wavelet transform, or DWT and statistical approaches, Logarithmic band power, or LBP, standard deviation, variance, or Shannon entropy retrieved features from the electroencephalogram (EEG) signals of the King Abdulaziz University (KAU) dataset. Four classifiers—k-nearest neighbor, linear discriminant analysis, support vector machine, and artificial neural networks (ANNs)—were fed these characteristics. According to the results, the optimal classifier combination is DWT+SE plus LBP. With SVM, the overall classification accuracy in the three-class single-channel mode reaches 99.9%. In contrast, with ANN, it drops to 97% (Alturki et al. 2020).
4. **Time–frequency domain features (T-FDF):** TFD, or time–frequency domain analysis, is an effective method for simultaneously investigating signals in two distinct time and frequency domains. Effectively capturing minute details and rapid EEG signal changes enables energy calculations in specified frequency and time ranges. There are two main types of TFD: discrete wavelet transform (DWT) and continuous wavelet transform (CWT). Several problems plague CWT, and DWT solves these issues by scaling and moving discretely. TFD is essential for deciphering complicated data such as EEG signals; however, it may be computationally demanding (Kim et al. 2020; ALYASSERI 2020; Singh and Krishnan 2023a, 2023b). (ALYASSERI 2020; Kim et al. 2020; Singh and Krishnan 2023a, 2023b).

According to recent studies, EEG Feature Extraction can help accurately classify individuals with ASD from those without it. One study used the wavelet transform to extract EEG rhythms from 29 subjects, 20 of whom had ASD and 9 were normal children. Following this, the participants were classified using pre-trained deep CNN models, which led to an F1-score of 99.19%, a sensitivity of 100%, and a specificity of 96.4% in automatically identifying ASD (Ari et al. 2022). Another recent study utilized a support vector machine (SVM) and DW) to categorize the EEG signals of ten children. Among them, five had low-functioning ASD (including two females and three males), and the other five had high-functioning ASD (including two females and five males). The study's results were spot on: 99.9% accuracy, 99.8% sensitivity, 100% specificity, and 100% precision (Divya et al. 2023) (Fig. 23).

Al-Qazzaz et al. (2021) The extracted EEG features must accurately depict the underlying brain components' neurophysiological activity. Therefore, it is crucial to recognize the BCI mental states correctly. Selecting a suitable collection of features impacts the filter classification performance more than choosing an effective classification model.

Autism spectrum disorder (ASD)-related brain abnormalities may be better detected with the use of EEG feature extraction. Wavelet transform is one approach to EEG feature extraction for autism diagnosis; it divides EEG signals into bands of varying frequencies and then extracts properties such as energy, entropy, and complexity from each band. This approach

may compare the power modulation and frequency distribution of individuals with ASD to those without the disorder and detect the non-stationary and transitory features of EEG data. That highlights how the two groups vary from one another (Mohi-Ud-Din and Jayanthi 2021c).

Entropy is a measure of the uncertainty or complexity of EEG signals. It can indicate the irregularity and unpredictability of EEG signals and differentiate between ASD and normal subjects regarding the amount of order and chaos in brain activity (Alhassan, Soudani, and Almusallam 2023).

A mathematical method known as the Fast Fourier Transform (FFT) may extract specific properties from electroencephalogram (EEG) data by converting them from the time domain to the frequency domain. Delta ranges from 0 to 4 Hz, theta from 4 to 8 Hz, alpha from 8 to 13 Hz, beta from 13 to 30 Hz, and gamma from 30 to 50 Hz comprise the output (Habeeb et al. 2019).

3.5 Use of machine learning and deep learning for autism spectrum disorder evaluation

More and more, researchers are turning to ML and DL methods to identify cases of autism spectrum disorder (ASD) (Bahathiq et al. 2022). Several metrics are used to assess the efficacy of these methods when applied. This section provides mathematical descriptions of many assessment metrics, the most popular of which is classification accuracy. Other metrics include the area under the curve (AUC), recall, F1, score, and accuracy. Table 4 displays the assessment metrics used to assess the outcomes of the ML and DL approaches to ASD categorization. Table 4 shows that all datasets apply at least one of these assessment metrics. As shown in Fig. 24, the most recent study employs over thirteen assessment met-

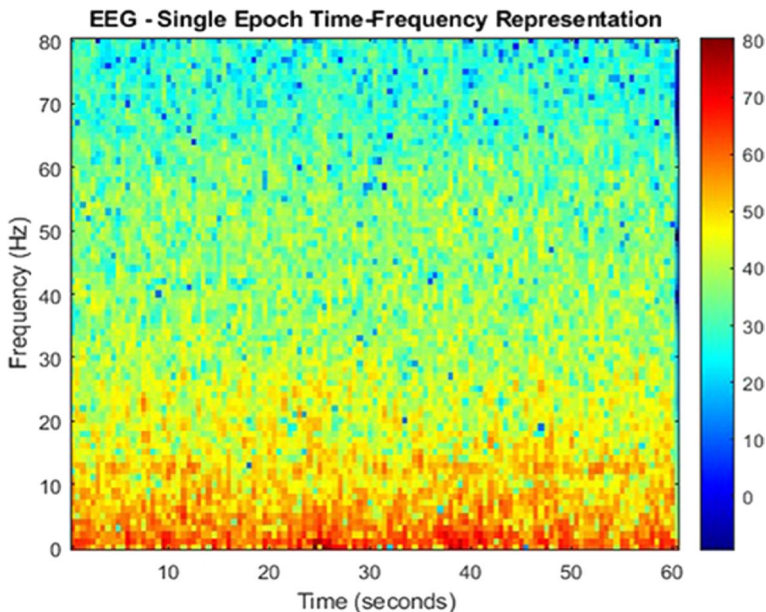


Fig. 23 Time–frequency representation of a signal using the S-transform (Singh and Krishnan 2023)

rics to assess ASD diagnostic models. To properly and comprehensively evaluate models, researchers require many assessment criteria. While sensitivity and specificity are often used, accuracy is the primary metric to assess their work. The percentage of actual positive results is called specificity, while the percentage of accurate negative results is called sensitivity. These metrics are crucial to finding out how well the model can distinguish between those with ASD and those with none. The three most popular methods for classifying data are k-nearest neighbors (kNN), neural networks (NN), and support vector machines (SVMs).

Table 5 compares some of the most used ML and DL models for the diagnosis of ASD. It shows the performance metrics considered for each model in recent works, including accuracy, sensitivity, and specificity. The comparison also shows how models such as CNN and SVM can be effective, presenting high accuracy and specificity during ASD diagnosis, as represented in Fig. 25.

4 Research gap and future works

Although EEG signals have shown success in diagnosing ASD with artificial intelligence (AI), several challenges remain to improve research quality. Here is a brief overview of these challenges:

1. **Heterogeneity and size samples:** One of the main challenges faced in studying ASD is the limited sample size, which may limit the generalizability and robustness of the results. The small sample size may have made it more susceptible to age effects, and therefore, any firm conclusions drawn from it may be questionable (Bahathiq et al. 2022; Aslam et al. 2022).
2. **Bias in datasets:** Another major challenge pertains to potential biases that might be part of the dataset on which training of the ML/DL models is done. Demographic inequalities due to age, gender, or ethnicity may affect performance. For example, ASD is more prevalent in males as compared to females; therefore, there may be a gender bias in the model. Similarly, poor representation of results from any particular ethnic group will reduce the generalization capabilities of the models. Because of these demographic disparities, future research must create more diverse datasets and develop ways to mitigate bias, such as data augmentation or weighting methods, to ensure model robustness across different demographic groups (Walensky et al. 2023; Siddique et al. 2024).
3. **Rehabilitation for autism:** Rehabilitation systems for ASD Excluded in most studies, autism rehabilitation includes various programs and services aimed at improving the quality of life and functioning of individuals with ASD. These interventions may include educational, behavioral, social, communication, sensory, motor, vocational programs, medication, and complementary therapies (Al-Qazzaz et al. 2021).
4. **Clinical applicability of the ML and DL models:** The big challenges of applying ML and DL models for ASD diagnosis in clinical practice are those concerning the generalizability and robustness of such models. While most studies report high accuracy in controlled research settings, this is further challenged in real-world clinical scenarios by variability in patient populations, equipment, and clinical protocols that apply in a given work environment. It can seriously affect the performance of these models.

Table 4 Evaluation measures for ASD diagnosis models

Measures	Equation	References
<i>Accuracy</i>	$Acc = \frac{Ta+Tr}{Ta+Fa+Tt+Fr} * 100$	(W.J. Bosl, Tager-Flusberg, and Nelson 2018a, 2018b; Ibrahim et al. 2018; Sudirman et al. 2018; Heunis et al. 2018; Abdulhay et al. 2018; Bajestani et al. 2019; Alturki et al. 2019; Aslam and Altaf 2019; Jayawardana, Jaime, and Jayarathna 2019; Haputhanthri et al. 2019b; Kumar et al. 2019; Grossi et al. 2019; Abdolzadegan et al. 2020; Kang et al. 2020; Bouallegue et al. 2020a; Zhao et al. 2020a, 2020b, 2020c; Majumder et al. 2020; Peya et al. 2020; Revathi et al. 2020; Alturki et al. 2020; Pham et al. 2020; Grossi et al. 2020; Zhao et al. 2020a, 2020b, 2020c; Haputhanthri et al. 2020; Vabalas et al. 2020; Sundaresan et al. 2021; Baygin et al. 2021; Dong et al. 2021; Mengucci et al. 2021; Rechal, Kumar, and Khaleelulla 2021; Aslam et al. 2021; Mohi-Ud-Din and Jayanthi 2021a; Oh et al. 2021; Tawhid et al. 2021a; Alotaibi and Maharatna 2021; Ranjani and Supraja 2021; Alturki et al. 2021; Mohi-Ud-Din and Jayanthi 2021b; Mohi Ud Din and Jayanthi 2022; Bakheet and Maharatna 2021; Wadhera and Kakkar 2021; Radhakrishnan et al. 2021; Ghoreishi et al. 2021; Ari et al. 2022; Liao, Duan, and Wang 2022a; Han et al. 2022; Yin et al. 2022; Wadhera and Mahmud 2022; Aslam et al. 2022; Alhassan and Soudani 2022; Thirumal and Thangakumar 2022; Sivasaravana Babu et al. 2023; Abdulhay et al. 2023; Divya et al. 2023; Tawhid et al. 2023; Ehwzhqh, Sdwlhqwv, and Khdownk 2023; Tang et al. 2023; Chen, Gaixiang, and Li 2023; Manoharan and Radhakrishnan 2023; Wadhera, Bedi, and Sharma 2023b; Visu et al. 2023; Alhassan, Soudani, and Almusallam 2023; Mayor Torres et al. 2023; Mohi Ud Din and Jayanthi 2023; Lakhan et al. 2023)
<i>Sensitivity</i>	$Sen (TAR, recall) = \frac{Ta}{Ta+Fr}$	(William J. Bosl et al. 2018b; Heunis et al. 2018; Abdulhay et al. 2018; Fan et al. 2018; Jayawardana, Jaime, and Jayarathna 2019; Kumar et al. 2019; Abdolzadegan et al. 2020; Bouallegue et al. 2020a; Pham et al. 2020; Grossi et al. 2020; Vabalas et al. 2020; Sundaresan et al. 2021; Baygin et al. 2021; Rechal, Kumar, and Khaleelulla 2021; Mohi-Ud-Din and Jayanthi 2021a; Oh et al. 2021; Tawhid et al. 2021a; Alotaibi and Maharatna 2021; Alturki et al. 2021; Bakheet and Maharatna 2021; Wadhera and Kakkar 2021; Radhakrishnan et al. 2021; Ghoreishi et al. 2021; Ari et al. 2022; J. Han et al. 2022; Yin et al. 2022; Alhassan and Soudani 2022; Wadhera and Mahmud 2022; Aslam et al. 2022; Thirumal and Thangakumar 2022; Sivasaravana Babu et al. 2023; Abdulhay et al. 2023; Divya et al. 2023; Tawhid et al. 2023; Manoharan and Radhakrishnan 2023; Visu et al. 2023; Alhassan, Soudani, and Almusallam 2023; Mohi Ud Din and Jayanthi 2023; Ehwzhqh, Sdwlhqwv, and Khdownk 2023; Wadhera, Bedi, and Sharma 2023a; Mayor Torres et al. 2023; Lakhan et al. 2023)

Table 4 (continued)

Measures	Equation	References
<i>Specificity</i>	$Spe(TFR) = \frac{Tr}{Tr+Fr}$	(Heunis et al. 2018; Abdulhay et al. 2018; Kumar et al. 2019; Bouallegue et al. 2020b; Pham et al. 2020; Grossi et al. 2020; Vabalas et al. 2020; Baygin et al. 2021; Rechal, Kumar, and Khaleelulla 2021; Mohi-Ud-Din and Jayanthi 2021a; Oh et al. 2021; Tawhid et al. 2021b; Alotaibi and Maharatna 2021; Alturki et al. 2021; Bakheet and Maharatna 2021; Wadhera and Kakkar 2021; Radhakrishnan et al. 2021; Ghoreishi et al. 2021; Ari et al. 2022; J. Han et al. 2022; Yin et al. 2022; Alhassan and Soudani 2022; Abdulhay et al. 2023; Divya et al. 2023; Tawhid et al. 2023; Manoharan and Radhakrishnan 2023; Visu et al. 2023; Alhassan, Soudani, and Almusallam 2023; Mohi Ud Din and Jayanthi 2023; Fan et al. 2018)
<i>F-Score</i>	$F-Score = 2 \times \left(\frac{Pre \times Recall}{Pre + Recall} \right)$	(Kumar et al. 2019; Rechal, Kumar, and Khaleelulla 2021; Baygin et al. 2021; Tawhid et al. 2021b; Ari et al. 2022; Yin et al. 2022; Wadhera and Mahmud 2022; Thirumal and Thangakumar 2022; Sivasaravana Babu et al. 2023; Tawhid et al. 2023; Ehwzhq, Sdwlhqw, and Khdownk 2023; Wadhera, Bedi, and Sharma 2023a; Alhassan, Soudani, and Almusallam 2023; Mayor Torres et al. 2023; Mengucci et al. 2021)
<i>Precision</i>	$Pre = \frac{Ta}{Ta+Fa}$	(Fan et al. 2018; Jayawardana, Jaime, and Jayarathna 2019; Kumar et al. 2019; Sundaresan et al. 2021; Baygin et al. 2021; Rechal, Kumar, and Khaleelulla 2021; Radhakrishnan et al. 2021; Wadhera and Mahmud 2022; Aslam et al. 2022; Thirumal and Thangakumar 2022; Mohi-Ud-Din and Jayanthi 2021a; Divya et al. 2023; Tawhid et al. 2023; Wadhera, Bedi, and Sharma 2023a; Mayor Torres et al. 2023; Mohi Ud Din and Jayanthi 2023; Lakhan et al. 2023)
<i>Receiver Operating Characteristic</i>	$ROC = \frac{TPR}{FPR}$	(Tawhid et al. 2021b; Alturki et al. 2021; Wadhera and Kakkar 2021; Han et al. 2022; Divya et al. 2023; Wadhera, Bedi, and Sharma 2023a; Visu et al. 2023; Kang et al. 2020)
<i>Area under curve</i>	$AUC = \int_b^a f(x) dx$	(Zhao et al. 2020a, 2020b, 2020c; Mengucci et al. 2021; Alturki et al. 2021; Yin et al. 2022; J. Han et al. 2022; Ehwzhq, Sdwlhqw, and Khdownk 2023; Wadhera, Bedi, and Sharma 2023a; Baygin et al. 2021)
<i>G_Mean</i>	$GM = \sqrt{Sen \times Spe}$	(Rechal, Kumar, and Khaleelulla 2021; Sudirman et al. 2018)
<i>Mean Square Error</i>	$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	(Manoharan and Radhakrishnan 2023; Jayawardana, Jaime, and Jayarathna 2019)
<i>Root Mean Square Error</i>	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{d_i - f_i}{\sigma_i} \right)^2}$	(Sheela and Puthankattil 2022; Abdulhay et al. 2018)

Table 4 (continued)

Measures	Equation	References
Positive Pre- dictive Value	$PPV = \frac{TP}{TP+FP}$	(Oh et al. 2021; Yin et al. 2022; Alhassan and Soudani 2022; Alhassan, Soudani, and Almusalam 2023; W.J. Bosl, Tager-Flusberg, and Nelson 2018a, 2018b)
Other (Mini- mum Mean Square Error (mMSE), prevalence Misclassifica- tion rate, and correlation coefficient)		(Jayawardana, Jaime, and Jayarathna 2019; Wadhera and Mahmud 2022; Bouallegue et al. 2020b; Mohi-Ud-Din and Jayanthi 2021a; Yin et al. 2022; Ehwzhhq, Sdwlhqvw, and Khdownk 2023)

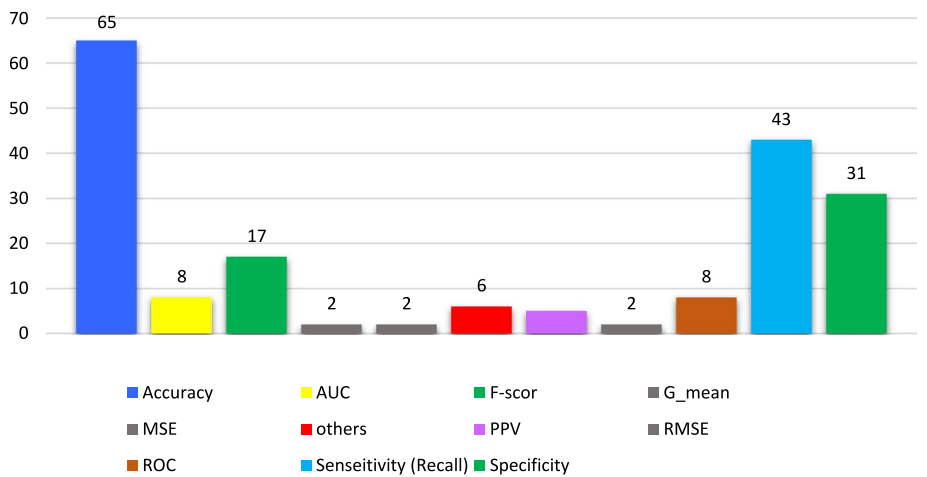
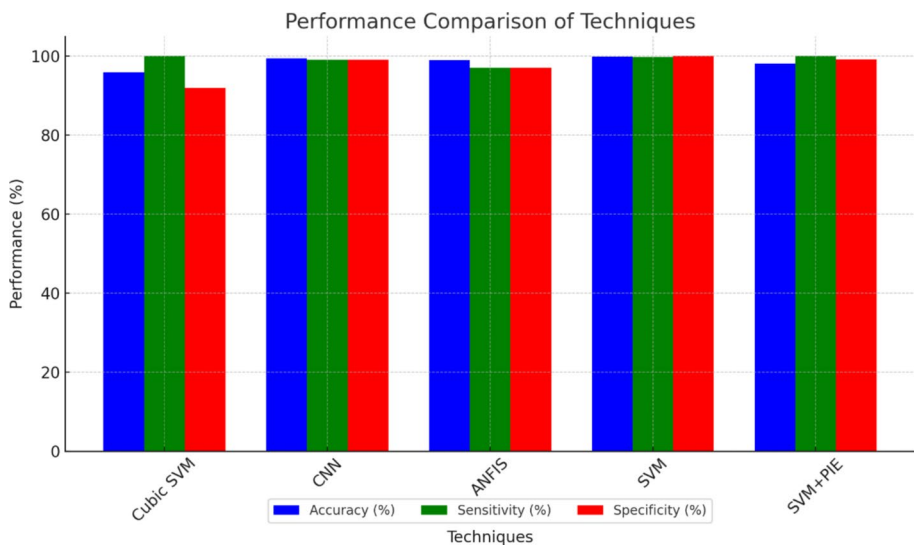


Fig. 24 Evaluation measures

Table 5 Performance comparison of ML and DL models in ASD diagnosis

Author	Techniques	Main Findings	Accuracy (%)	Sensitivity (%)	Specificity (%)
Kumar et al. (2019)	ANFIS	Precision of 97%, accuracy of 98.9%	98.9	97	97
Rechal, Kumar, and Khaleelulla (2021)	SVM+PIE	The fusion of SVM and PIE achieved an accuracy of 98.08	98.08	100	99.16
Alotaibi and Maharatra (2021)	Cubic-SVM	95.8% accuracy, 100% sensitivity	95.8	100	92
Tawhid et al. (2023)	CNN	CNN achieved 99.39% accuracy	99.39	99	99
Divya et al. (2023)	SVM	99.9% accuracy, 100% Specificity	99.9	99.8	100

**Fig. 25** Accuracy, Sensitivity, and Specificity Comparison of ML/DL Models for ASD Diagnosis

Furthermore, clinical environments often lack the large, diverse datasets required to train these models effectively (Paneru et al. 2024; Liao, Duan, and Wang 2022b).

- AI model systems into clinical workflows:** Other most important challenges involve integrating these AI models into clinical workflows. Clinicians must have confidence in these model predictions and be informed of the underlying logic behind the decision-making of these models. The AI systems must be transparent, interpretable, and easy

to use. For this, more clinical trials need to be carried out to help validate the effectiveness of such models in realistic clinical settings. Most importantly, collaboration among researchers, clinicians, and healthcare institutions is called for seamlessly integrating such technologies into daily clinical practice (Nasarian et al. 2024).

6. **Ethical considerations in using ML/DL for ASD diagnosis:** Many critical ethical issues arise in work about diagnosing ASD using ML and DL models. Data privacy becomes one of the foremost considerations; such models demand sensitive health information, such as EEG recordings. If the data is secured and kept confidential, it will help maintain the patients' trust. Any breach of privacy would seriously raise questions about adopting AI systems in healthcare. Besides, the ML/DL models may be susceptible to mistakes, leading to misdiagnosis, inappropriate treatment, or delayed care. Rigorous testing and clinical validation will be obligatory for these models before they can be fitted clinically. The second major ethical consideration is demographic biases in the datasets in which these models were trained. Namely, underrepresentation in the case of females or ethnic minorities may result in biased predictions made by the models, further increasing healthcare disparities. Therefore, future research should develop fair and transparent models tested across diverse populations to ensure equity in healthcare outcomes. (Byrne 2018; Kumar, Rai, and Pandey 2018; Sreerama and Krishnamoorthy 2022; Sharma and Sharma 2024).
7. **End-to-end system:** No real-time system fully utilizes real-time data. Only the state-of-the-art (Lakhan et al. 2023) presented real-time functions in IoT applications for ASD patients, showing efficiency and learning capabilities based on a trained dataset.
8. **Best features:** There are different features; no one has compared the features and chosen the best feature.
9. There are different types and levels of ASD; no studies have focused on classifying its types and levels.

This review paper suggests potential for further research in this area. By studying ASD from different angles, researchers can gain new insights and contribute to understanding of the disorder such as:

1. **Use of Large Datasets and Rigorous Methodologies:** It is crucial to use large datasets and rigorous research methodologies to ensure that findings can be applied to various individuals. AI is expected to improve significantly shortly, which could aid in diagnosing ASD and provide valuable support to clinicians. Moreover, to create accurate predictive models, having a diverse sample, including individuals from different backgrounds, ethnicities, races, and income groups, is essential. Large and diverse sample sizes are necessary for accurate results. Collaborating across samples is crucial for early predictive models.
2. **Diversity in Dataset:** Developing larger datasets but, at the same time, diversity will be an important task for future research to reduce biases and improve the accuracy of ASD diagnosis ML/DL models. It should capture various demographic variations such as age, gender, and ethnicity. Diversity and representativeness in data ensure applicability across different populations in general.

3. **Feature Identification for Rehabilitation:** Understanding and identifying features within EEG can improve the diagnosis of ASD and rehabilitation processes through specific treatment.
4. **Generalizable models:** Researchers should develop more generalizable models that could be applied in various clinical settings. It might be achievable by adopting data augmentation, transfer learning, and training models on heterogeneous datasets to make models more robust and adaptable.
5. **Clinical Practice Collaboration:** Integrating AI-based diagnostic tools into clinical practice requires collaboration between AI researchers, healthcare professionals and policymakers. The emphasis should be on developing scientifically sound, practical, and user-friendly models for everyday use within clinical settings.
6. **Clinical Practice Engagement:** AI researchers, healthcare professionals, and policymakers must collaborate more to insert AI-based diagnostic tools into clinical practice. The aim is to secure scientifically sound but also practical and user-friendly models for clinicians' daily workflows.
7. **Specificity of the findings:** They suggest that further testing across different clinical populations and age groups is needed to understand the potential of EEG to predict comorbidities and functional outcomes.
8. **Feature identification for rehabilitation:** Future research should pursue feature identification within EEG data. It can help diagnose ASD and the rehabilitation process, leading to targeted and effective treatment.
9. **Real-Time Systems:** Real-time systems are likely to provide faster and more convenient screening and assessment for ASDs, especially for remote and underserved populations. These systems should reduce human error and bias, facilitate continuous monitoring, provide feedback on symptoms of ASD, and provide intervention strategies for optimizing the quality of care and patient outcomes.
10. **Work on classifying the types and levels of ASD.**
11. **Improved Data-Extraction Features and Model Interpretability:** Future work should be done to gather more comprehensive data, refine feature extraction and selection methods, and develop reliable and interpretable ML/DL models. This provides helpful information for better understanding and creating suitable models for clinical applications.

5 Conclusions

This paper is a comprehensive analysis of autism spectrum disorder (ASD) techniques used to diagnose and recognize it, which is increasing in prevalence. This work primarily aims to synthesize prior research relevant to ASD and summarise it. The examined works have been chosen from many public and prestigious academic sources, including Elsevier, IEEE, MDPI, and Springer. Selecting these publications involves using many filters to exclude redundant concepts about autism and use EEG and feature extraction techniques. Typically, the research on ASD is examined and deliberated on three main categories: machine learning, deep learning, and hybrid approaches. Machine learning research is categorized based on classifiers such as NNs, ANNs, PNNs, SVM, KNN, and others, as well as other methods.

Overall, SVM is extensively used for diagnosing, classifying, and predicting ASD, whereas CNN is the predominant deep-learning method in this domain.

Moreover, the research about deep learning is classified as CNN, RNN, DNN, and Optimization techniques for deep learning. Most of the studies summarised in this analysis used EEG signals to diagnose ASD, with some also incorporating eye-tracking data. These studies utilized primary or secondary datasets from authors previously used them for ASD diagnosis. While most datasets are unavailable online, their accessibility is detailed in Table 1. to available data or not. A request for a dataset from the author or institution is required, taking into account the conditions to be used by other researchers to evaluate and compare their methods. The preprocessing stage involves applying filters to raw EEG data, such as the passband, low pass, and high pass filters. These filters are used to enhance data quality and reduce noise.

Additionally, EEG Feature Extraction can assist in accurately distinguishing individuals with ASD from those without it. This process involves identifying relevant and discriminative information from EEG signals. It is crucial to select the best feature for diagnosing ASD. The assessment metrics used to diagnose ASD using machine learning and deep learning techniques are outlined. The accuracy, sensitivity, and specificity metrics have been extensively used in prior investigations. That aligns with previous research that used deep learning and machine learning approaches for the prediction procedure.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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Authors and Affiliations

Hajir Ammar Hatim^{1,4} · Zaid Abdi Alkareem Alyasseri^{1,2,3} · Norziana Jamil⁵

✉ Zaid Abdi Alkareem Alyasseri
Zaid.alyasseri@uokufa.edu.iq; zaid.alyasseri@ukm.edu.my

✉ Norziana Jamil
Norziana@uacu.ac.ac
Hajir Ammar Hatim
hajara.alzubaydi@uokufa.edu.iq

- ¹ Information Technology Research and Development Center (ITRDC), University of Kufa, Najaf, Iraq
- ² Center for Artificial Intelligence Technology, Faculty of Information Science and Technology, Universiti Kebangsaan Malaysia, Bangi, Malaysia
- ³ College of Engineering, University of Warith Al-Anbiyaa, Karbala, Iraq
- ⁴ Department of Computer Science, Faculty of Education, University of Kufa, Najaf, Iraq
- ⁵ College of IT, United Arab Emirates University, Al Ain, PO Box 15551, Abu Dhabi, UAE